EDA and MCMC Methods for Babies Dataset

Yatziri I. Carmona Ochoa

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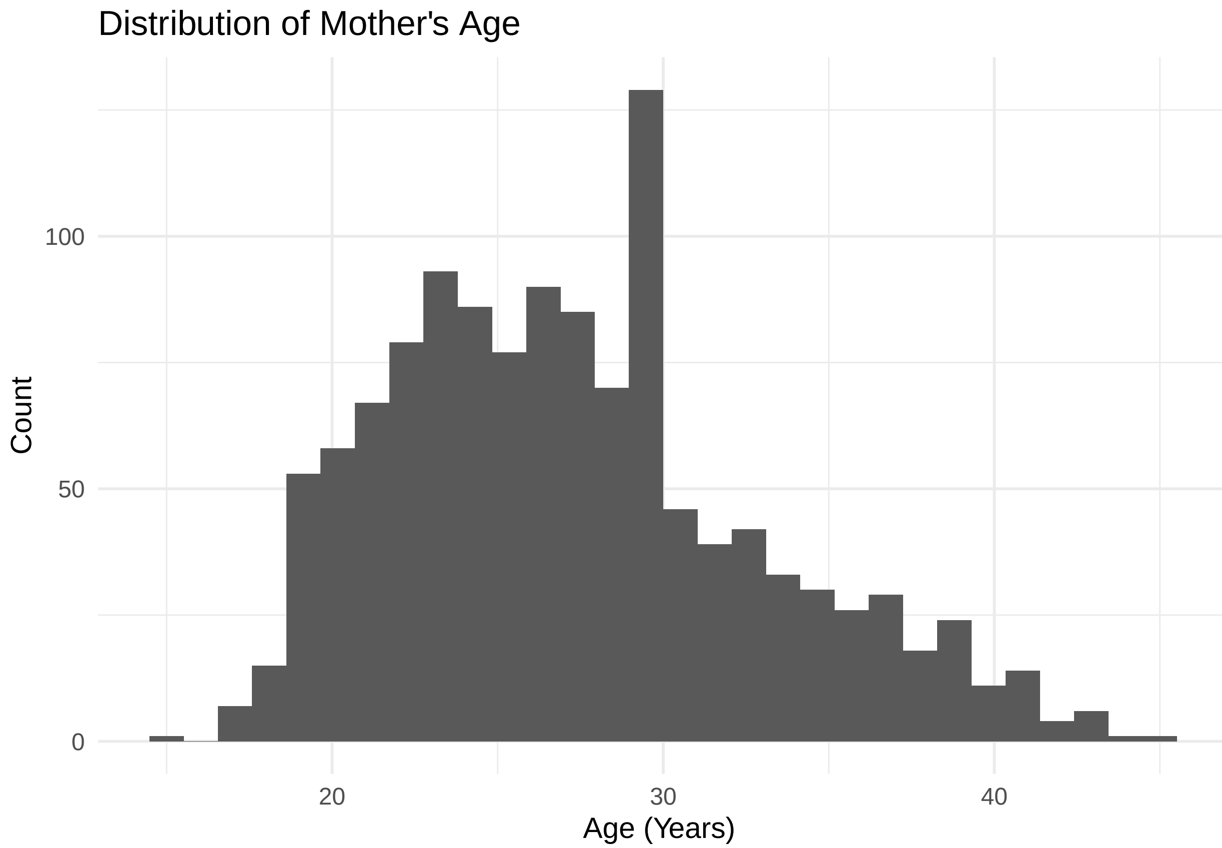
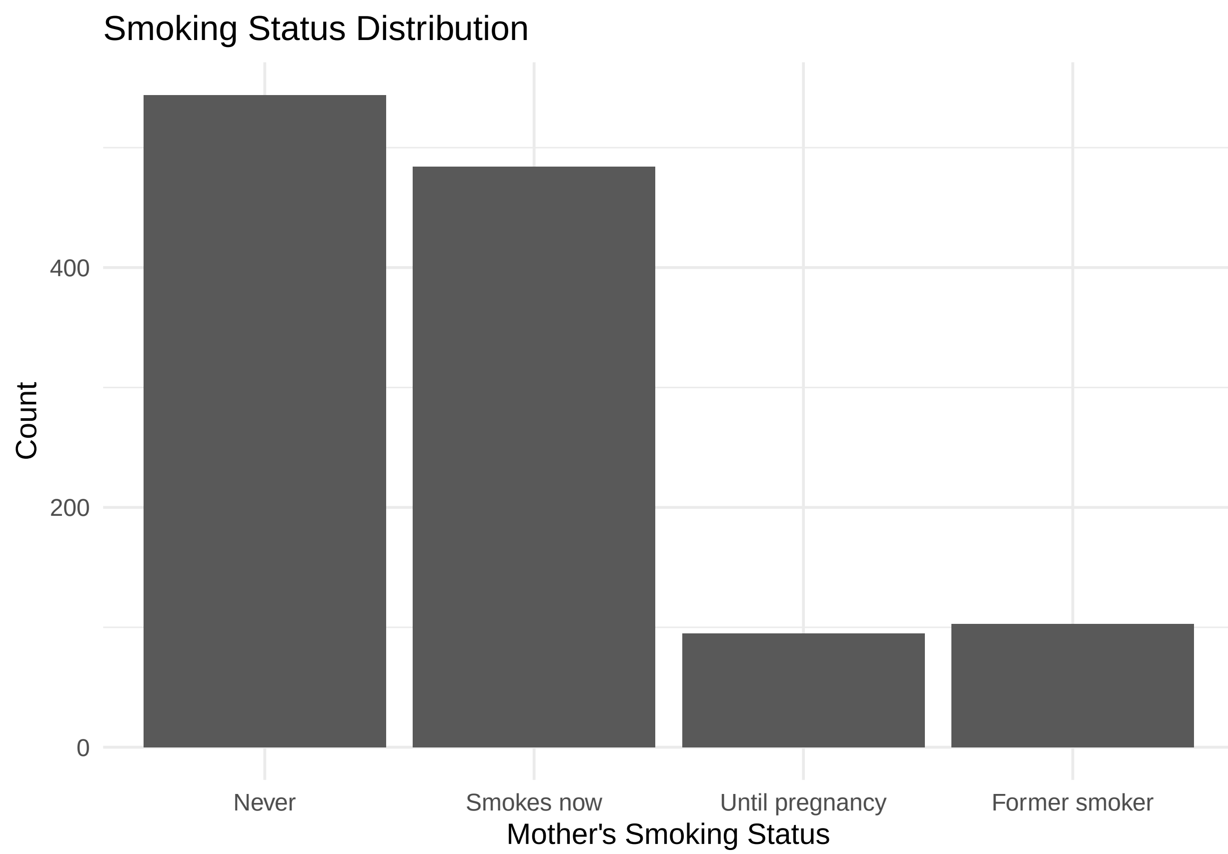
```r
library(UsingR)
library(Bolstad)
library(ggplot2)
library(reshape2)
library(ggridges)
library(dplyr)
```

EDA

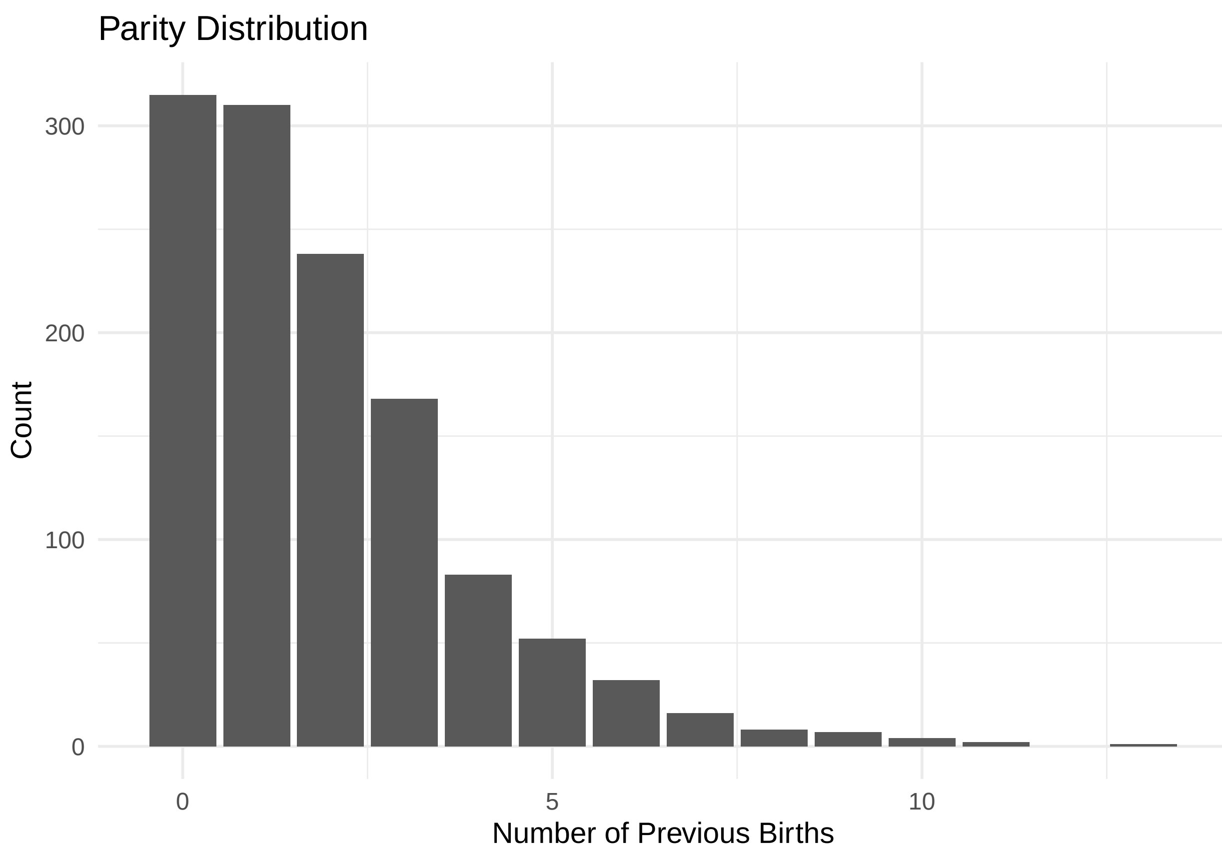
```r
babies_clean <- babies

#removing missing/incomplete data
babies_clean[babies_clean == 999] <- NA
babies_clean$age[babies_clean$age == 99] <- NA
babies_clean$wt1[babies_clean$wt1 == 999] <- NA
babies_clean$smoke[babies_clean$smoke == 9] <- NA
babies_clean$race[babies_clean$race == 99] <- NA
babies_clean$inc[babies_clean$inc == 99] <- NA
babies_clean$inc[babies_clean$inc == 98] <- NA
babies_clean$sex[babies_clean$sex == 9] <- NA
babies_clean$ed[babies_clean$ed == 9] <- NA
babies_clean$ht[babies_clean$ht == 99] <- NA
babies_clean$parity[babies_clean$parity == 99] <- NA

#age distribution
ggplot(babies_clean, aes(x = age)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of Mother's Age",
       x = "Age (Years) ", y = "Count") +
  theme_minimal()
```



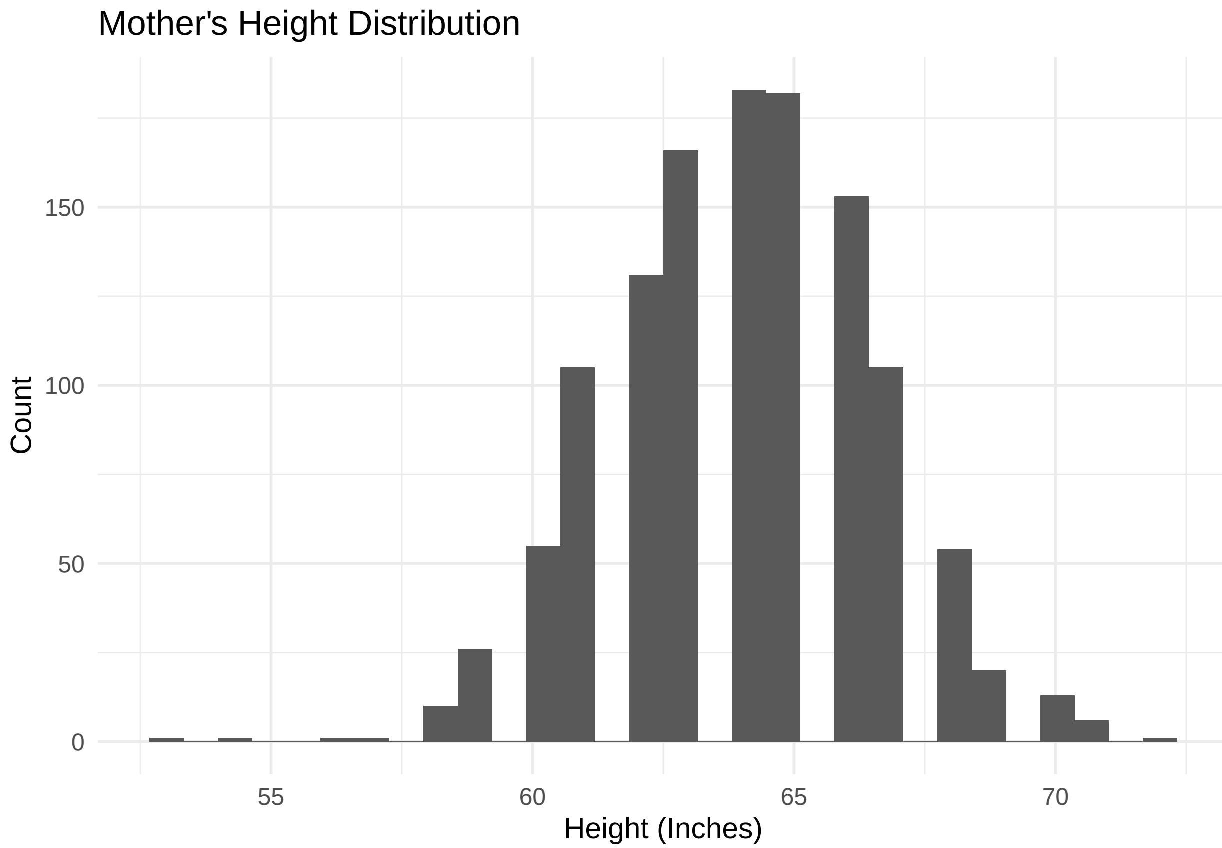

```r
#smoking status distribution
babies_clean$smoke_group <- with(babies_clean,
  ifelse(smoke == 0, "Never",
  ifelse(smoke == 1, "Smokes now",
  ifelse(smoke == 2, "Until pregnancy",
  ifelse(smoke == 3, "Former smoker", NA)))))

babies_clean$smoke_group <- factor(
  babies_clean$smoke_group,
  levels = c("Never", "Smokes now", "Until pregnancy", "Former smoker")
)

ggplot(subset(babies_clean, !is.na(smoke_group)), aes(x = smoke_group)) +
  geom_bar() +
  labs(
    title = "Smoking Status Distribution",
    x = "Mother's Smoking Status",
    y = "Count"
  ) +
  theme_minimal()
```



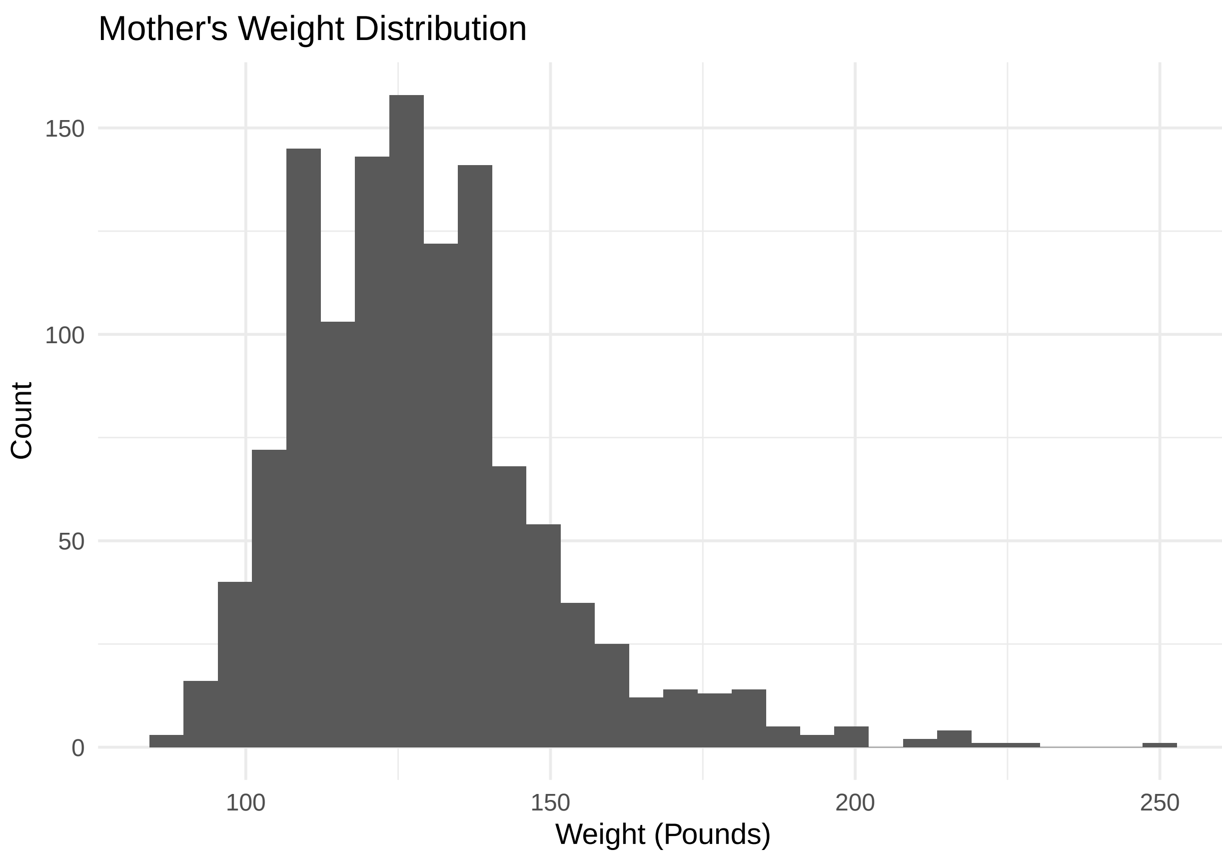

```r
#parity distribution
ggplot(babies_clean, aes(x = parity)) +
  geom_bar() +
  labs(title = "Parity Distribution",
       x = "Number of Previous Births", y = "Count") +
  theme_minimal()
```



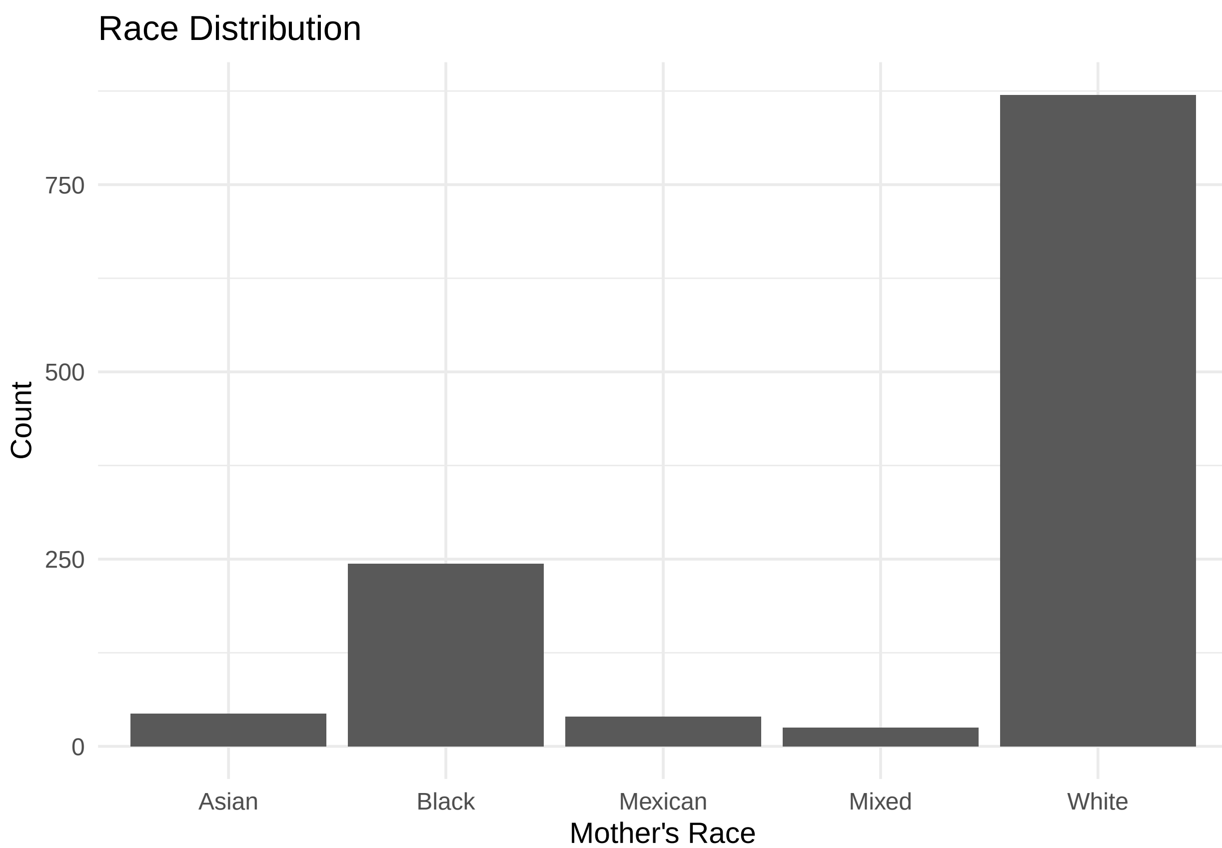

```r
#height distribution for moms
ggplot(babies_clean, aes(x = ht)) +
  geom_histogram(bins = 30) +
  labs(title = "Mother's Height Distribution",
       x = "Height (Inches)", y = "Count") +
  theme_minimal()
```




```r
#mother's weight distribution
ggplot(babies_clean, aes(x = wt1)) +
  geom_histogram(bins = 30) +
  labs(title = "Mother's Weight Distribution",
       x = "Weight (Pounds)", y = "Count") +
  theme_minimal()
```

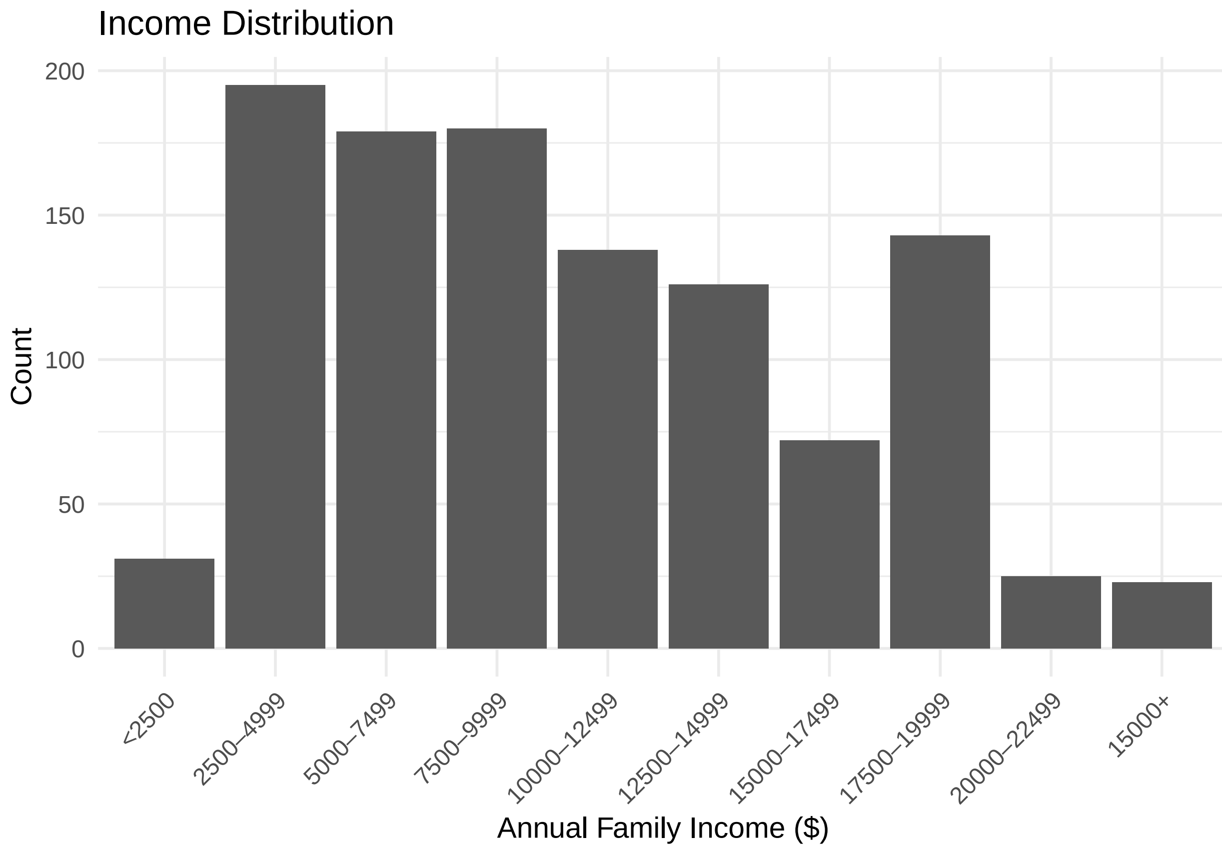



```
#mother's race distribution
babies_clean$race_group <- with(babies_clean,
  ifelse(race %in% 0:5, "White",
  ifelse(race == 6, "Mexican",
  ifelse(race == 7, "Black",
  ifelse(race == 8, "Asian",
  ifelse(race == 9, "Mixed", NA))))))

babies_clean$race_group[babies_clean$race == 10] <- NA

ggplot(subset(babies_clean, !is.na(race_group)), aes(x = race_group)) +
  geom_bar() +
  labs(
    title = "Race Distribution",
    x = "Mother's Race",
    y = "Count"
  ) +
  theme_minimal()
```

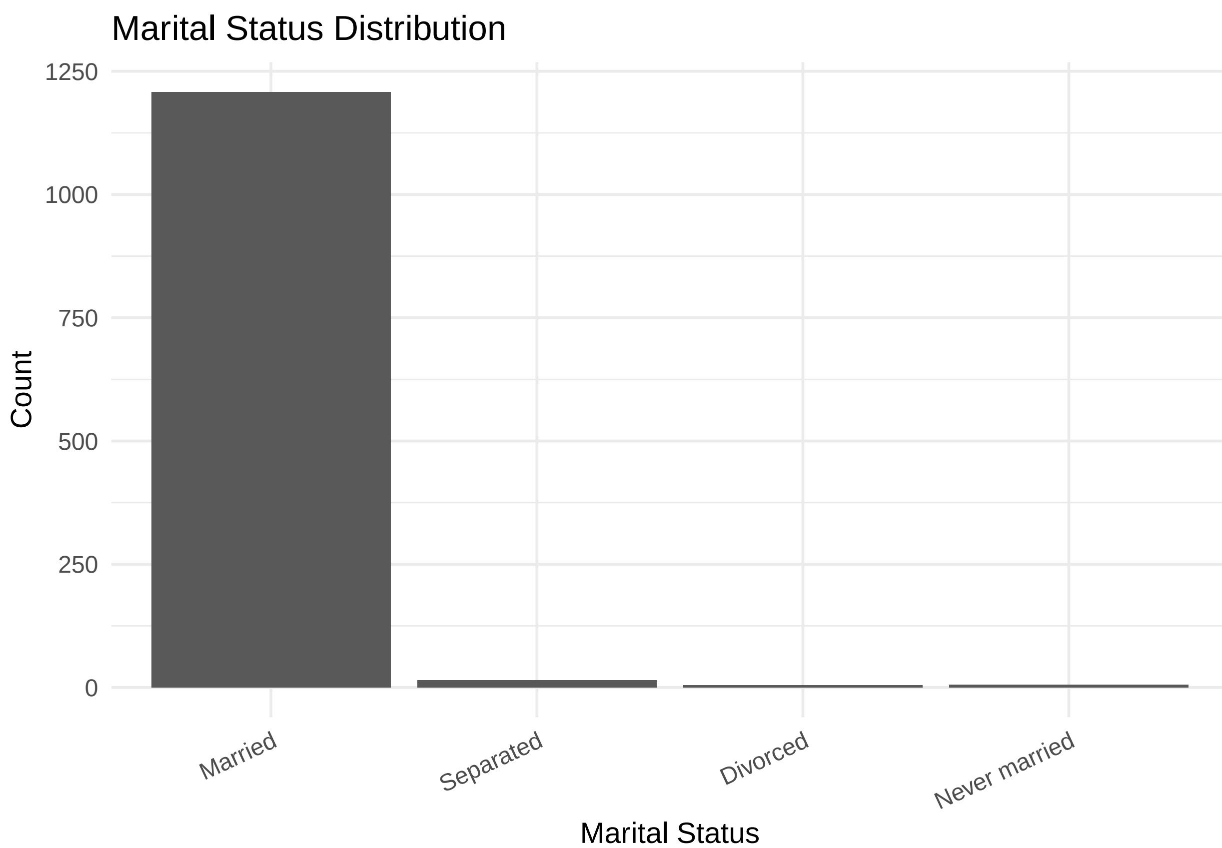



```r
#annual income
babies_clean$inc_group <- factor(
  babies_clean$inc,
  levels = 0:9,
  labels = c(
    "<2500",
    "2500-4999",
    "5000-7499",
    "7500-9999",
    "10000-12499",
    "12500-14999",
    "15000-17499",
    "17500-19999",
    "20000-22499",
    "15000+"
  )
)

ggplot(subset(babies_clean, !is.na(inc_group)), aes(x = inc_group)) +
  geom_bar() +
  labs(
    title = "Income Distribution",
    x = "Annual Family Income ($)",
    y = "Count"
  ) +
  theme_minimal() +
```

```
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

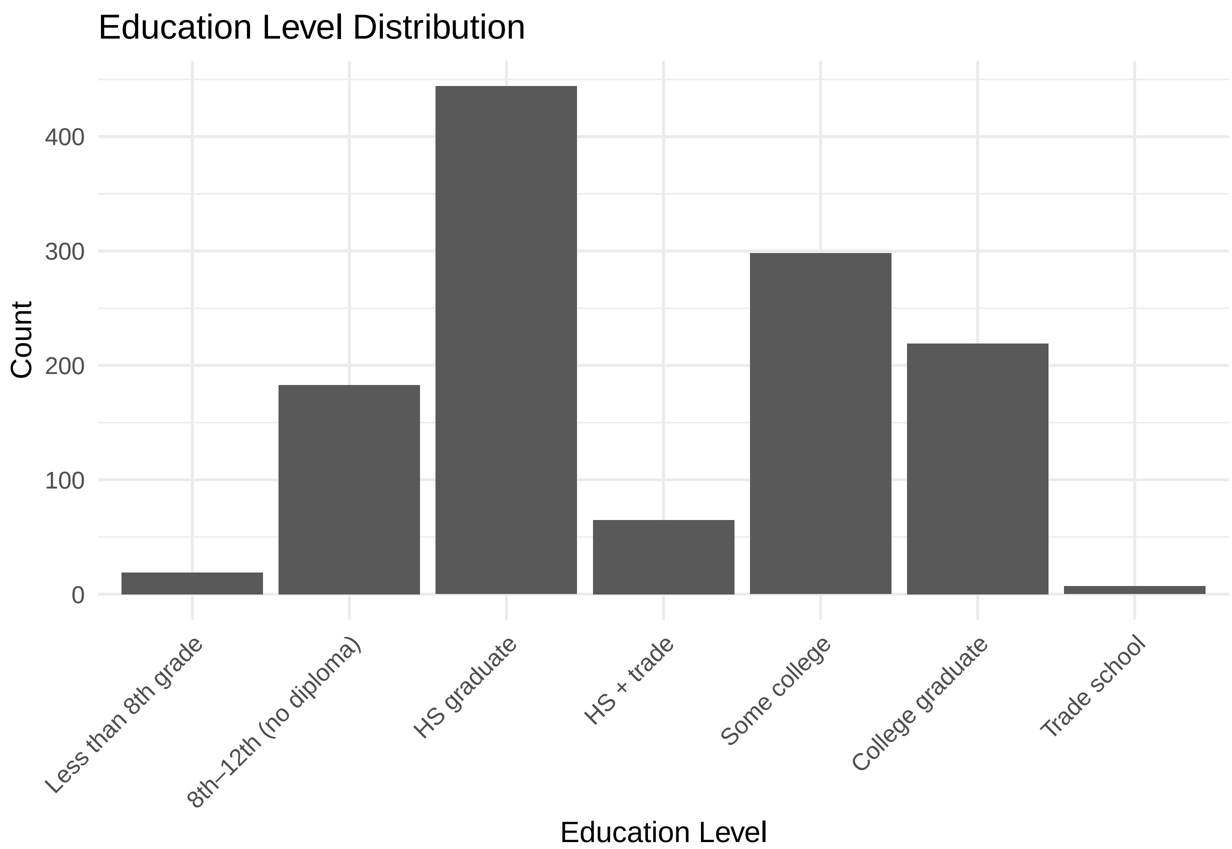



```
#marital status distribution
babies_clean$marital_group <- factor(
  babies_clean$marital,
  levels = c(1,2,3,4,5),
  labels = c(
    "Married",
    "Separated",
    "Divorced",
    "Widowed",
    "Never married"
  )
)

ggplot(subset(babies_clean, !is.na(marital_group)), aes(x = marital_group)) +
  geom_bar() +
  labs(
    title = "Marital Status Distribution",
    x = "Marital Status",
    y = "Count"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 25, hjust = 1))
```

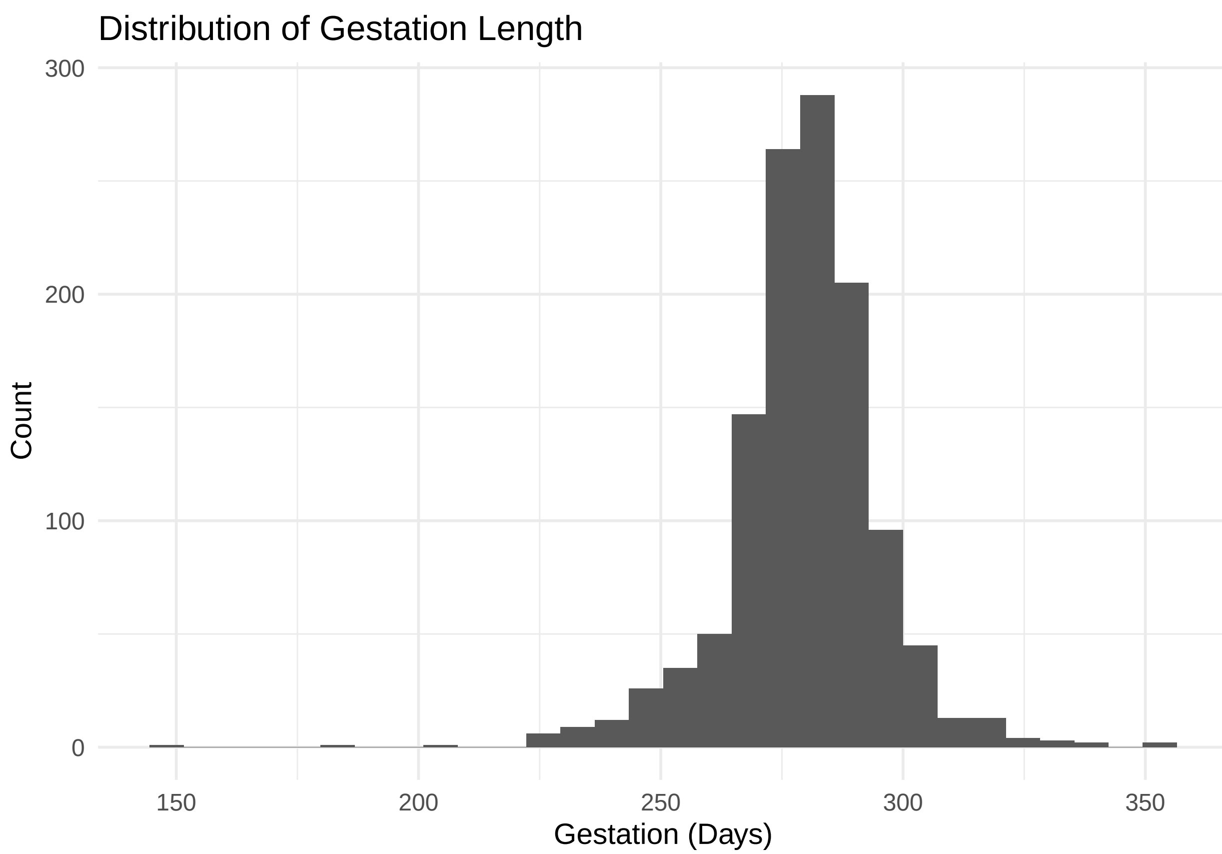



```
#mother's education distribution
babies_clean$edu_group <- with(babies_clean,
  ifelse(ed == 0, "Less than 8th grade",
  ifelse(ed == 1, "8th-12th (no diploma)",
  ifelse(ed == 2, "HS graduate",
  ifelse(ed == 3, "HS + trade",
  ifelse(ed == 4, "Some college",
  ifelse(ed == 5, "College graduate",
  ifelse(ed %in% c(6,7), "Trade school", NA))))))))

babies_clean$edu_group <- factor(
  babies_clean$edu_group,
  levels = c(
    "Less than 8th grade",
    "8th-12th (no diploma)",
    "HS graduate",
    "HS + trade",
    "Some college",
    "College graduate",
    "Trade school"
  )
)

ggplot(subset(babies_clean, !is.na(edu_group)), aes(x = edu_group)) +
  geom_bar() +
  labs(
```

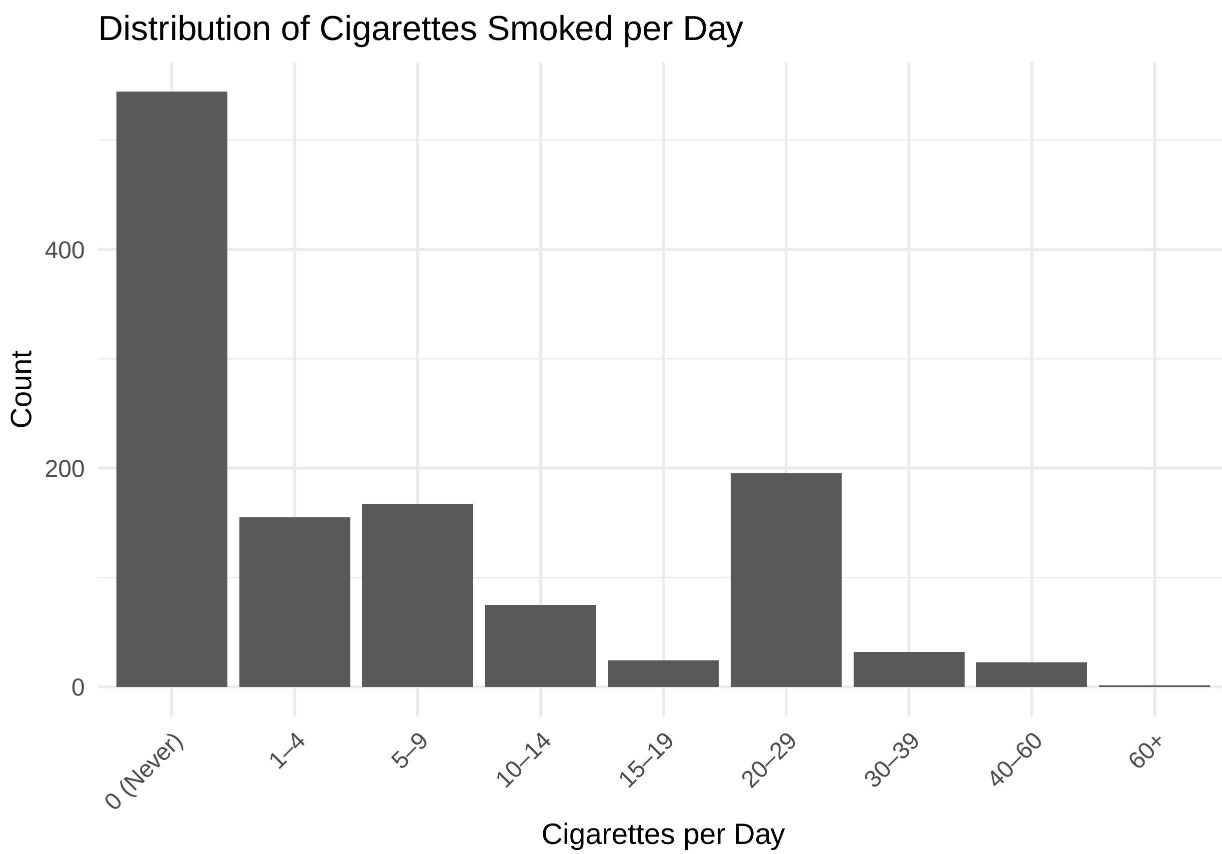
```r
    title = "Education Level Distribution",
    x = "Education Level",
    y = "Count"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```r
#gestation
ggplot(babies_clean, aes(x = gestation)) +
  geom_histogram(bins = 30) +
  labs(
    title = "Distribution of Gestation Length",
    x = "Gestation (Days)",
    y = "Count"
  ) +
  theme_minimal()
```




```r
#number of cigarettes smoked
babies_clean$cigs_cat <- factor(
  babies_clean$number,
  levels = 0:8,
  labels = c(
    "0 (Never)",
    "1-4",
    "5-9",
    "10-14",
    "15-19",
    "20-29",
    "30-39",
    "40-60",
    "60+"
  )
)

ggplot(subset(babies_clean, !is.na(cigs_cat)), aes(x = cigs_cat)) +
  geom_bar() +
  labs(
    title = "Distribution of Cigarettes Smoked per Day",
    x = "Cigarettes per Day",
    y = "Count"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



#EDA Proportions, Counts, Range Etc

```r
#age
range(babies_clean$age, na.rm = TRUE)
```

```
## [1] 15 45
```

```r
#smoke status
smoke_summary <- babies_clean %>%
  count(smoke_group) %>%
  mutate(prop = n / sum(n))
smoke_summary
```

```
##       smoke_group   n        prop
## 1           Never 544 0.440129450
## 2      Smokes now 484 0.391585761
## 3 Until pregnancy  95 0.076860841
## 4   Former smoker 103 0.083333333
## 5            <NA>  10 0.008090615
```

```r
#race
race_summary <- babies_clean %>%
  count(race_group) %>%
  mutate(prop = n / sum(n))
race_summary
```

```
##   race_group   n       prop
## 1      Asian  44 0.03559871
## 2      Black 244 0.19741100
## 3    Mexican  40 0.03236246
## 4      Mixed  25 0.02022654
## 5      White 870 0.70388350
## 6       <NA>  13 0.01051780
```

```r
#height
range(babies_clean$ht, na.rm = TRUE)
```

```
## [1] 53 72
```

```r
#weight
range(babies_clean$wt1, na.rm = TRUE)
```

```
## [1]  87 250
```

```r
#parity
range(babies_clean$parity, na.rm = TRUE)
```

```
## [1]  0 13
```

```r
#gestation
range(babies_clean$gestation, na.rm = TRUE)
```

```
## [1] 148 353
```

```r
#cigs
cigs_summary <- babies_clean %>%
  filter(!is.na(cigs_cat)) %>%
  count(cigs_cat) %>%
  mutate(prop = n / sum(n))
cigs_summary
```

```
##    cigs_cat   n         prop
## 1 0 (Never) 544 0.4477366255
## 2       1-4 155 0.1275720165
## 3       5-9 167 0.1374485597
## 4     10-14  75 0.0617283951
## 5     15-19  24 0.0197530864
## 6     20-29 195 0.1604938272
## 7     30-39  32 0.0263374486
## 8     40-60  22 0.0181069959
## 9       60+   1 0.0008230453
```

```r
#education
edu_summary <- babies_clean %>%
  filter(!is.na(edu_group)) %>%
  count(edu_group) %>%
  mutate(prop = n / sum(n))
edu_summary
```

```
##                edu_group   n        prop
## 1   Less than 8th grade  19 0.015384615
## 2 8th-12th (no diploma) 183 0.148178138
## 3           HS graduate 444 0.359514170
## 4            HS + trade  65 0.052631579
## 5          Some college 298 0.241295547
## 6      College graduate 219 0.177327935
## 7          Trade school   7 0.005668016
```

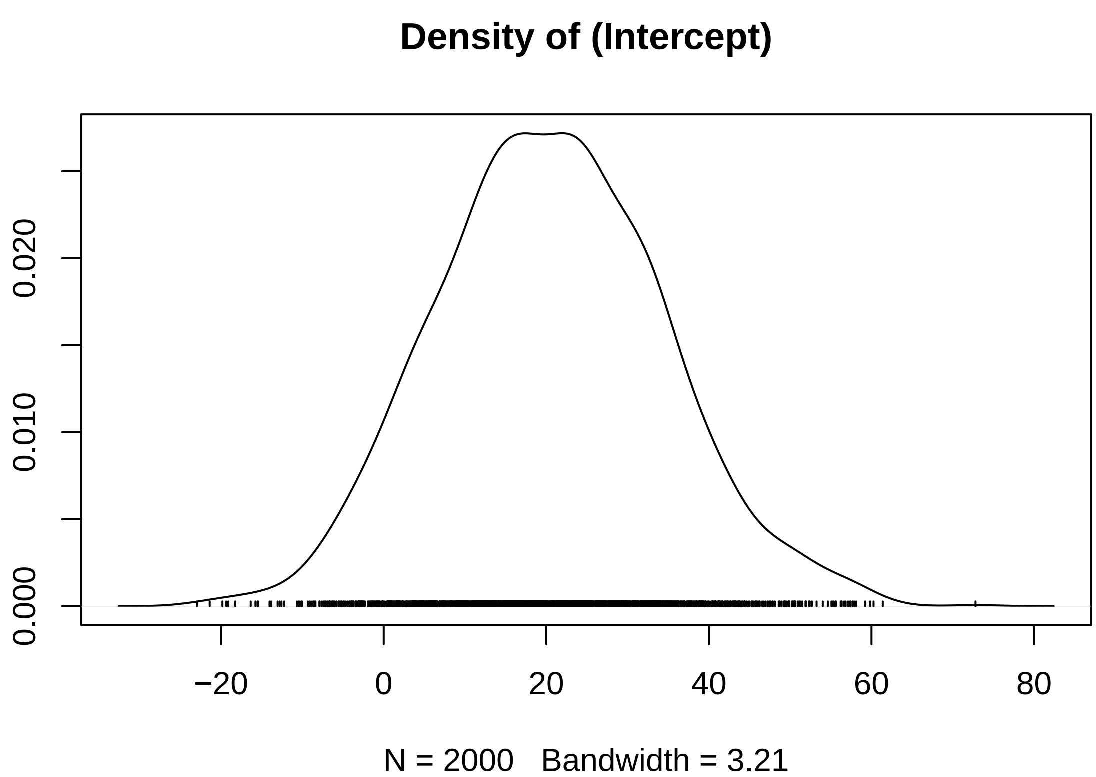
MCMC for Bayesian Linear Regression Non-Informed

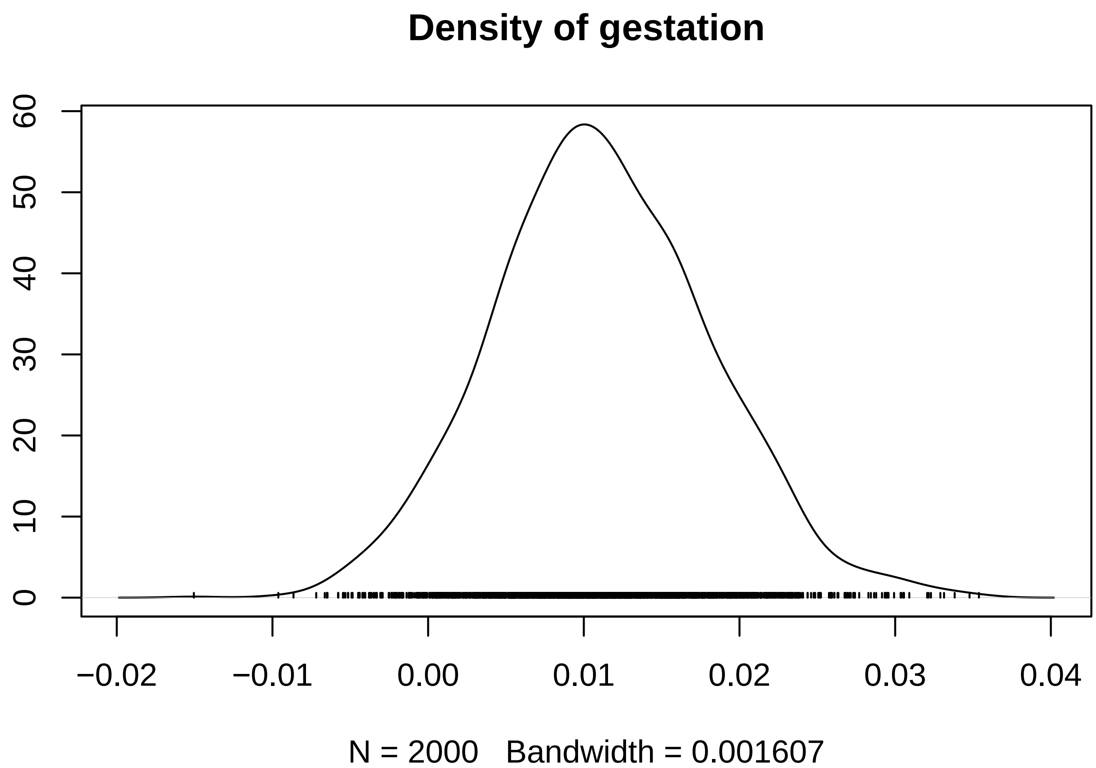
```r
library(MCMCpack)

babies_df2 <- babies

#clean dataset
babies_clean_2 <- subset(babies_df2,
  wt != 999 &
  parity != 99 &
  age != 99 &
  ht != 99 &
  wt1 != 999 &
  smoke != 9 &
  !(number %in% c(9, 98, 99))
)

model <- MCMCregress(
  wt ~ gestation + ht + wt1 + number + parity + age,
  data = babies_clean_2,
  burnin = 2000,
  mcmc = 10000,
  thin = 5
)
summary(model)
```

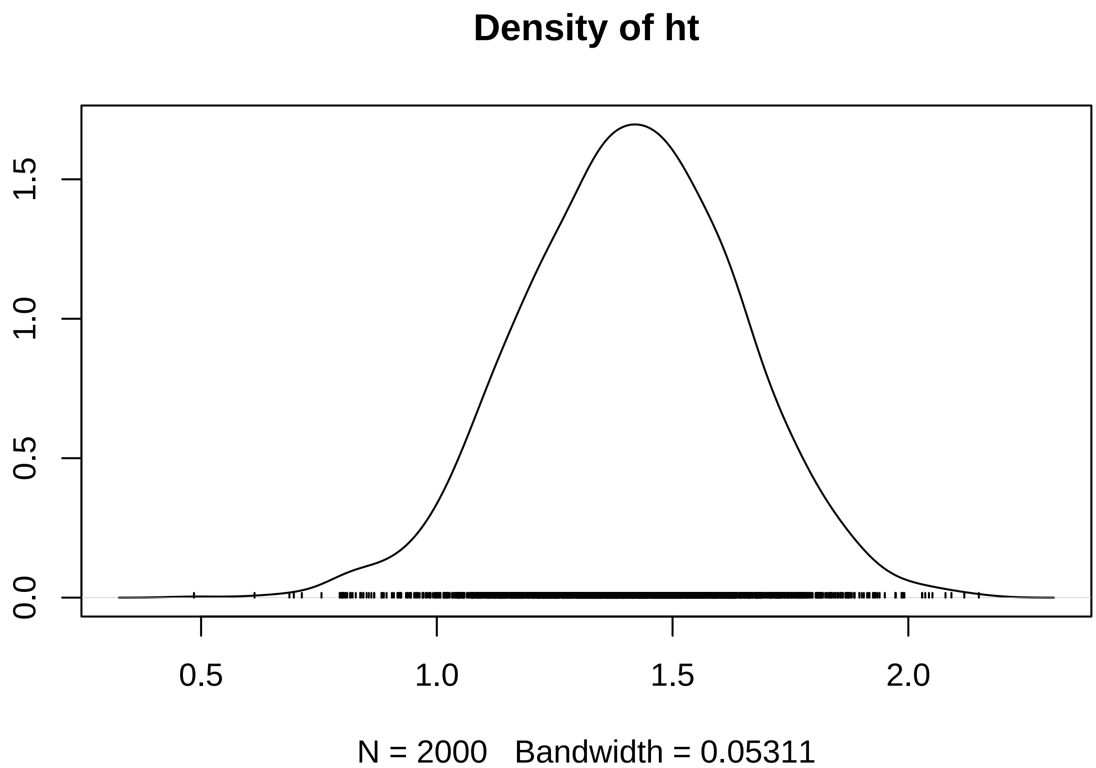
```
##
## Iterations = 2001:11996
## Thinning interval = 5
## Number of chains = 1
## Sample size per chain = 2000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##                 Mean        SD  Naive SE Time-series SE
## (Intercept)  20.46387 13.847975 0.3096501       0.279934
## gestation     0.01113  0.006971 0.0001559       0.000161
## ht            1.41042  0.229107 0.0051230       0.004737
## wt1           0.05711  0.028310 0.0006330       0.000633
## number       -1.79584  0.248489 0.0055564       0.005556
## parity        0.06451  0.312991 0.0069987       0.006999
## age           0.03757  0.102709 0.0022966       0.002209
```

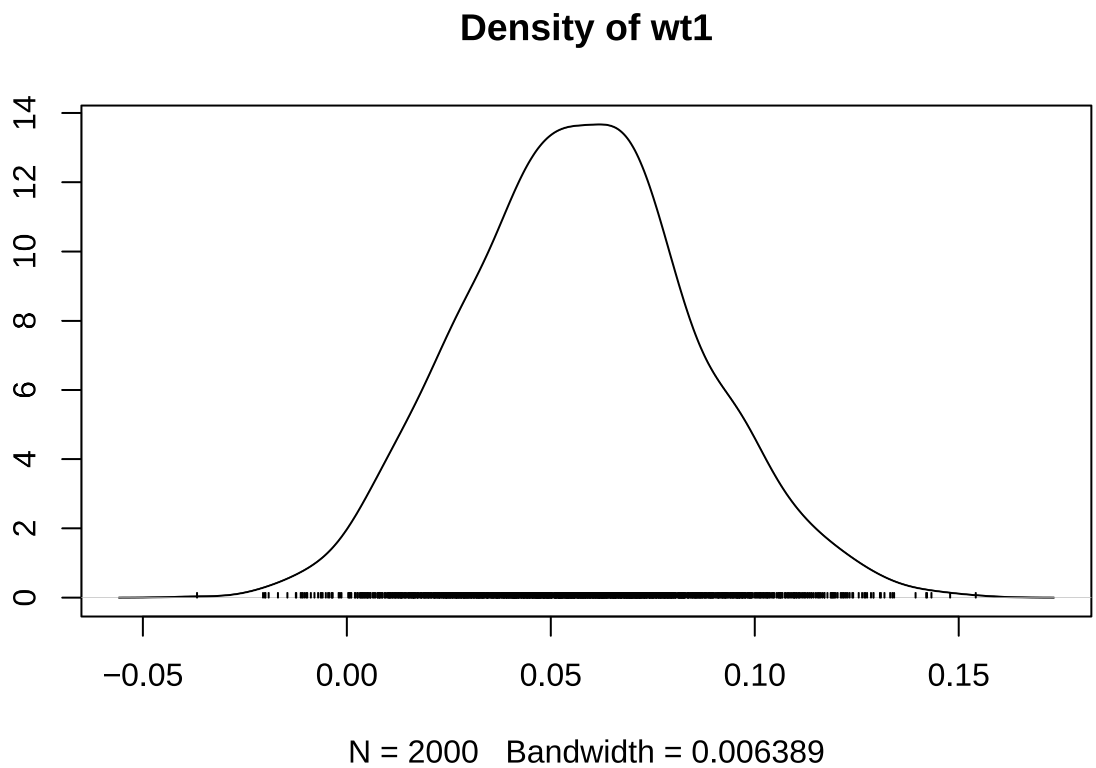
```
## sigma2      308.95856 12.962438 0.2898489       0.289849
##
## 2. Quantiles for each variable:
##
##                   2.5%        25%       50%       75%     97.5%
## (Intercept)  -5.504138  11.195160  20.19066  29.93518  49.55231
## gestation    -0.002068   0.006377   0.01084   0.01566   0.02513
## ht            0.955084   1.253830   1.41478   1.56675   1.85136
## wt1           0.003847   0.038243   0.05675   0.07518   0.11491
## number       -2.284053  -1.956842  -1.79795  -1.63088  -1.31222
## parity       -0.535379  -0.142989   0.05901   0.26994   0.69138
## age          -0.163235  -0.030945   0.03701   0.10590   0.23722
## sigma2      285.342717 300.116901 308.64709 317.22553 335.74883
```

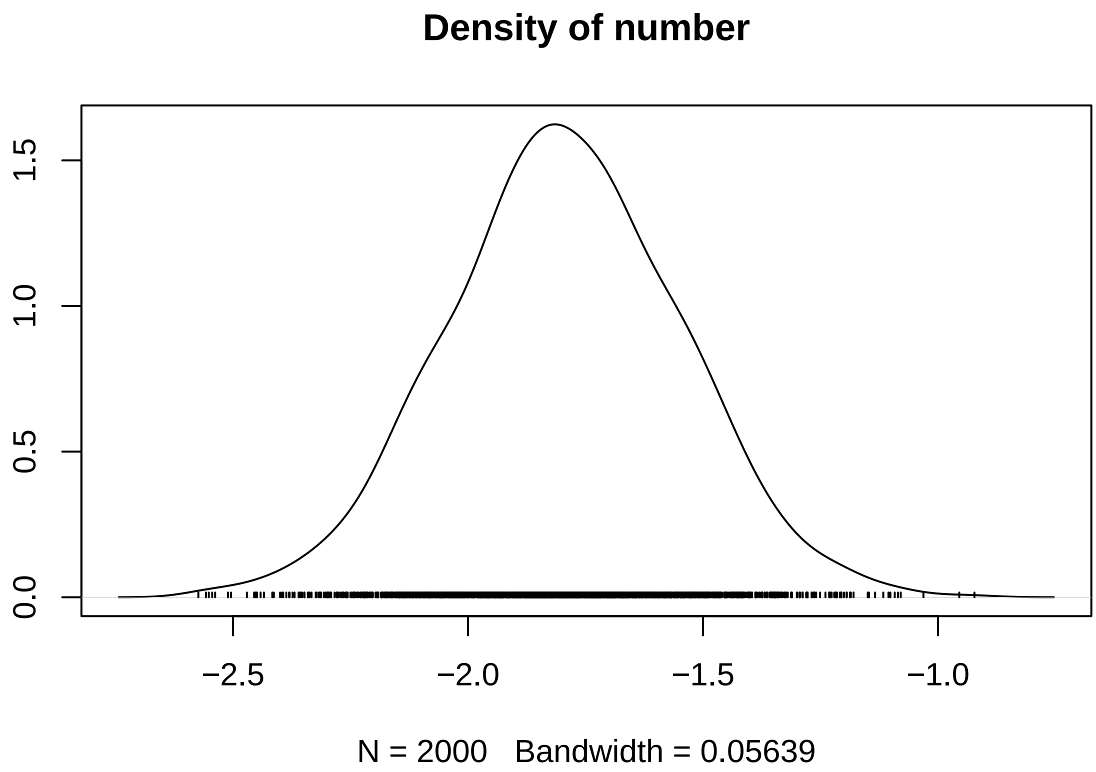
```r
densplot(model)
```

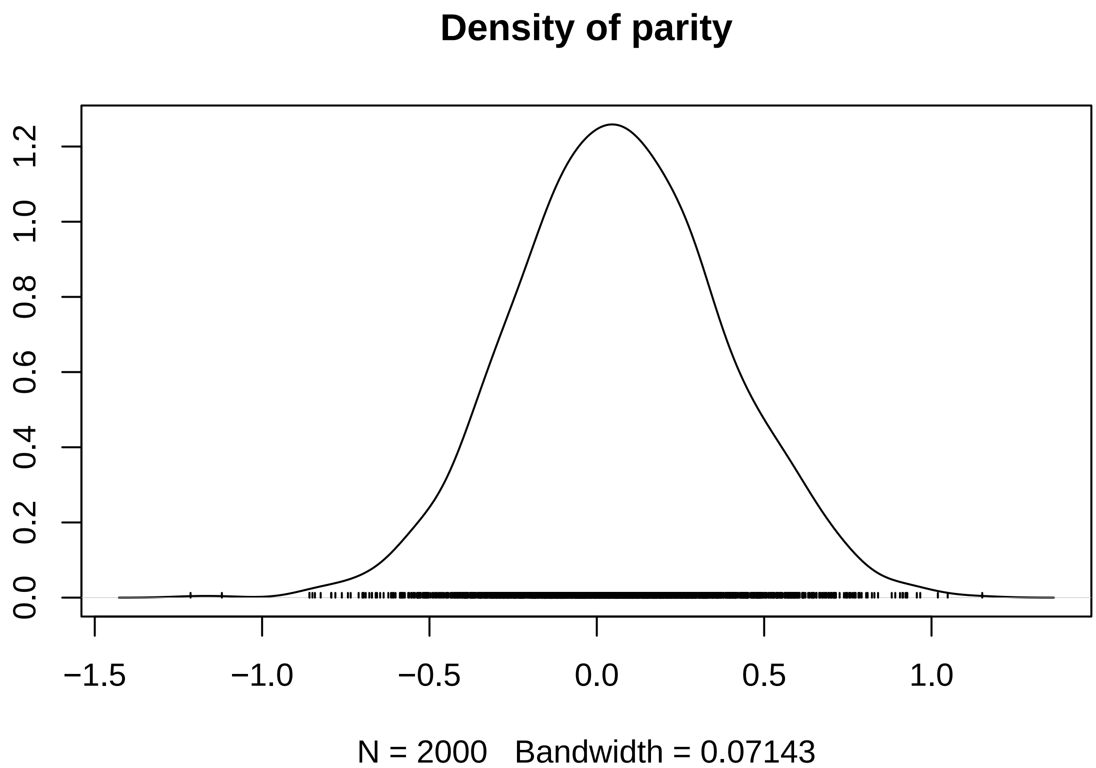


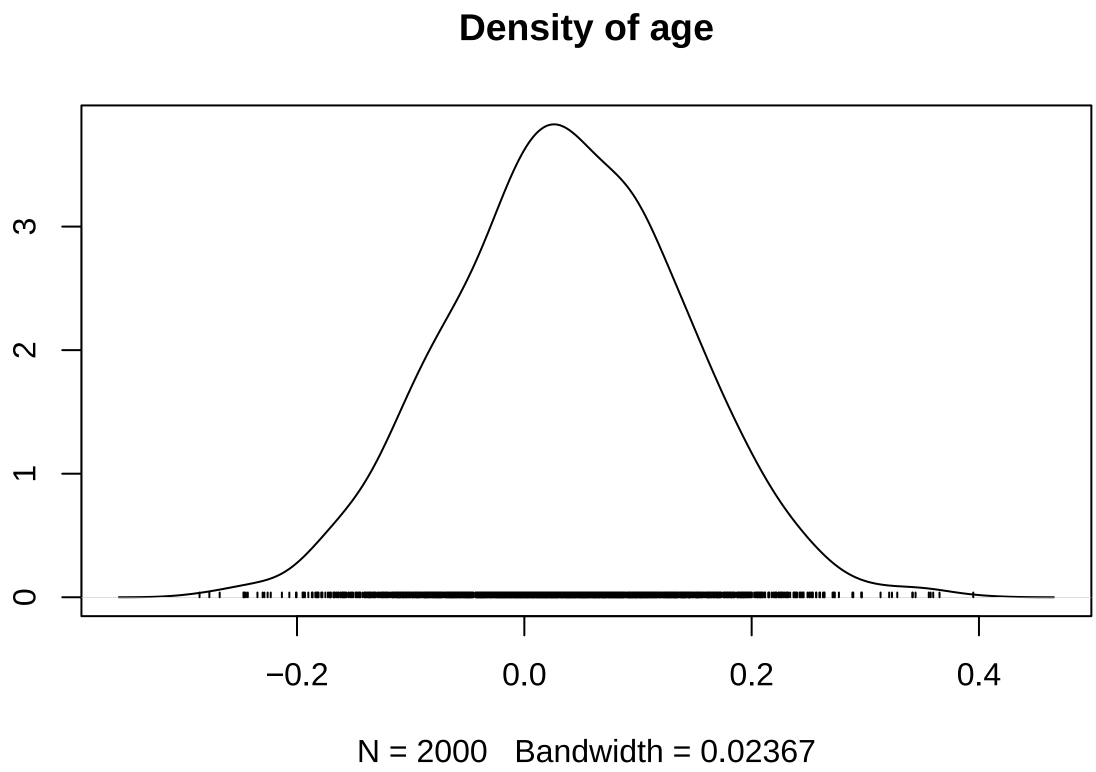

Density of gestation
0
10
20
30
40
50
60
−0.02
−0.01
0.00
0.01
0.02
0.03
0.04
N = 2000 Bandwidth = 0.001607

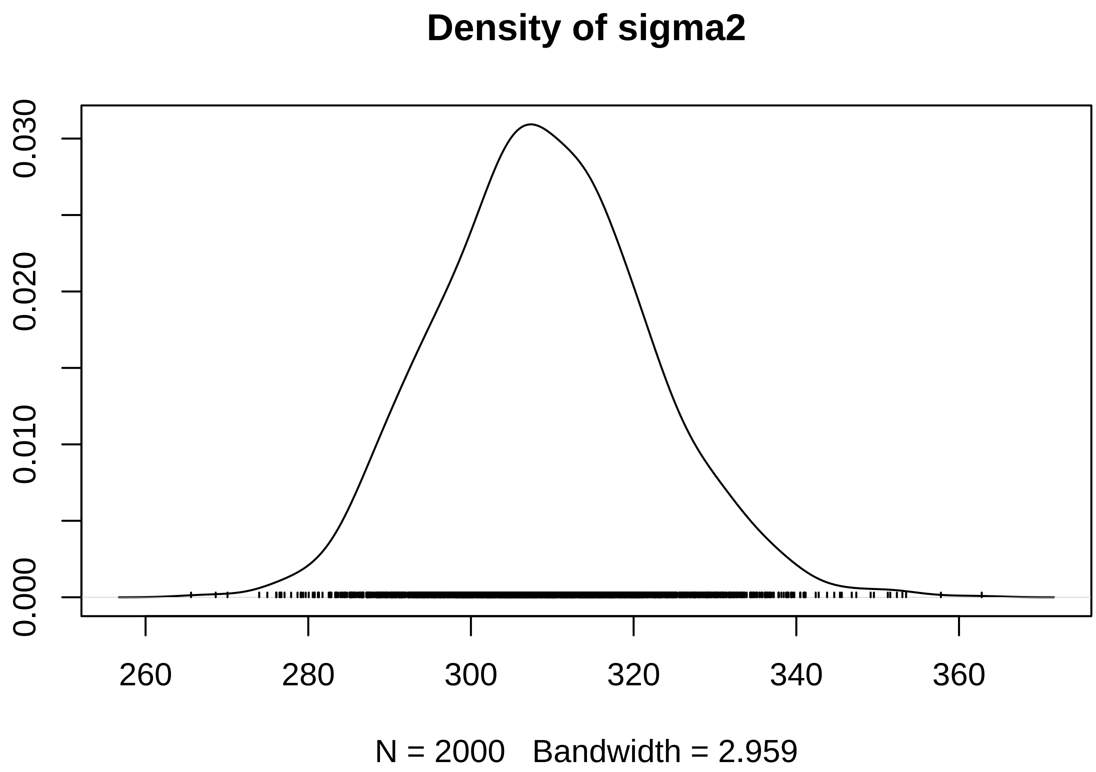

Density of ht
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
N = 2000   Bandwidth = 0.05311


Density of wt1
0
2
4
6
8
10
12
14
−0.05
0.00
0.05
0.10
0.15
N = 2000 Bandwidth = 0.006389

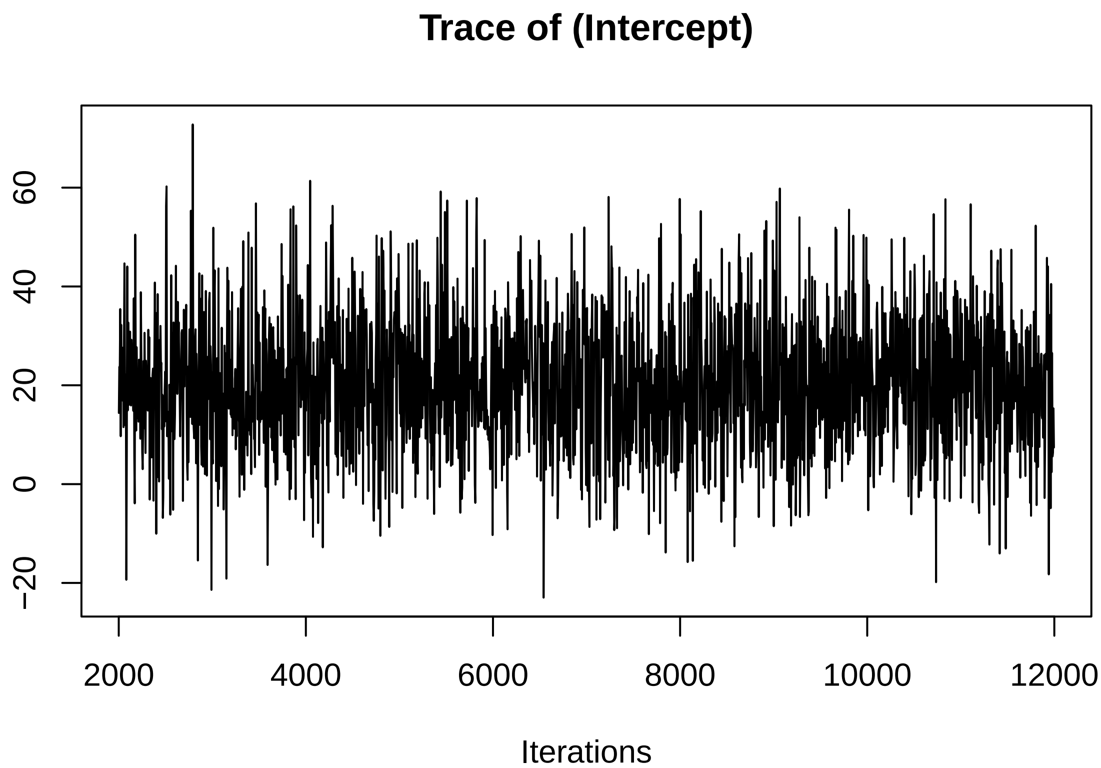

Density of number
0.0
0.5
1.0
1.5
−2.5
−2.0
−1.5
−1.0
N = 2000 Bandwidth = 0.05639

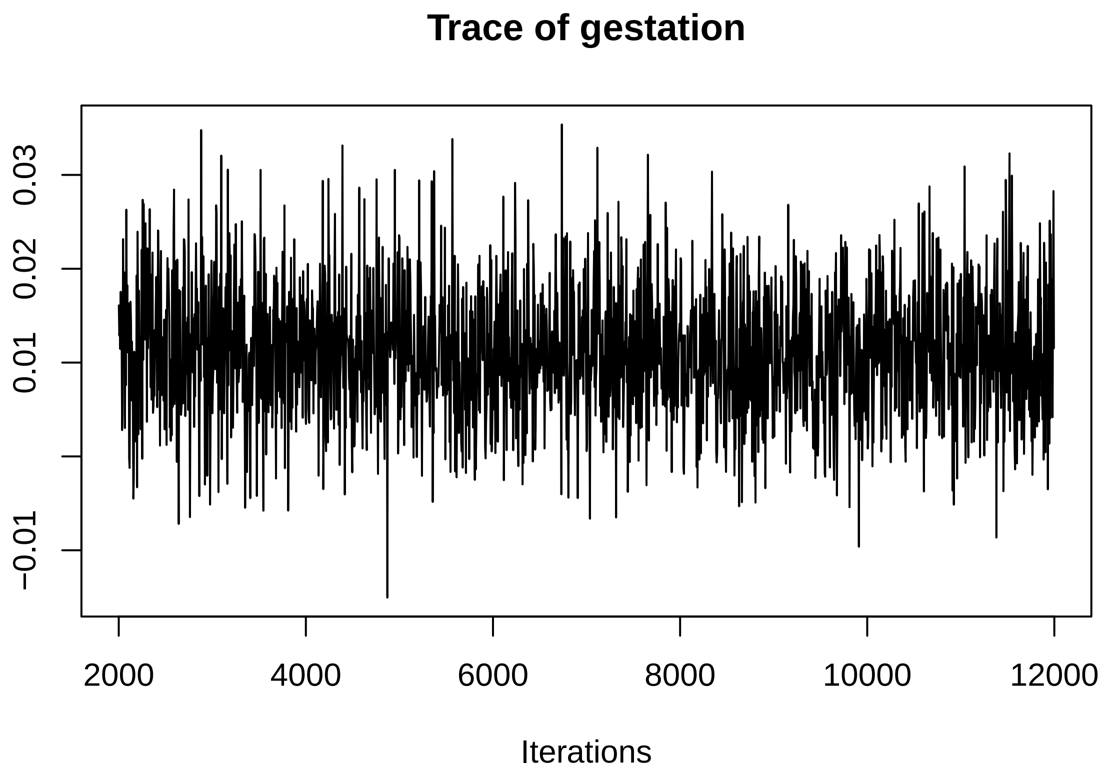

Density of parity
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
N = 2000 Bandwidth = 0.07143

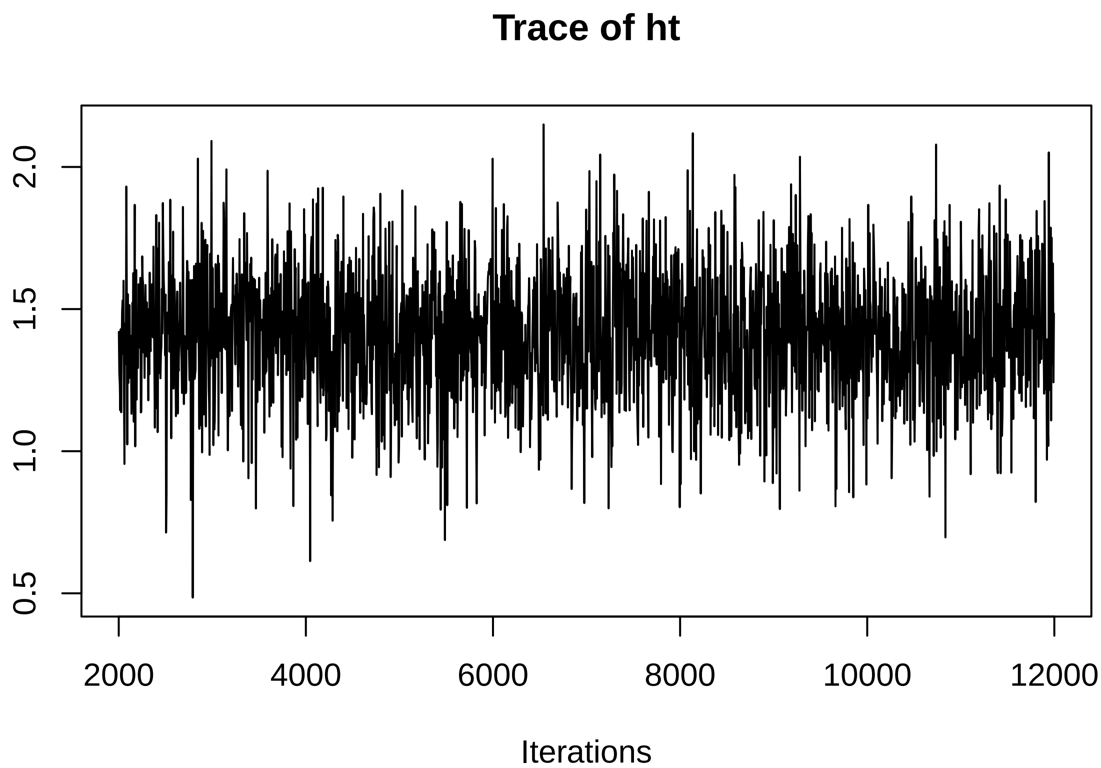

Density of age
0
1
2
3
−0.2
0.0
0.2
0.4
N = 2000   Bandwidth = 0.02367

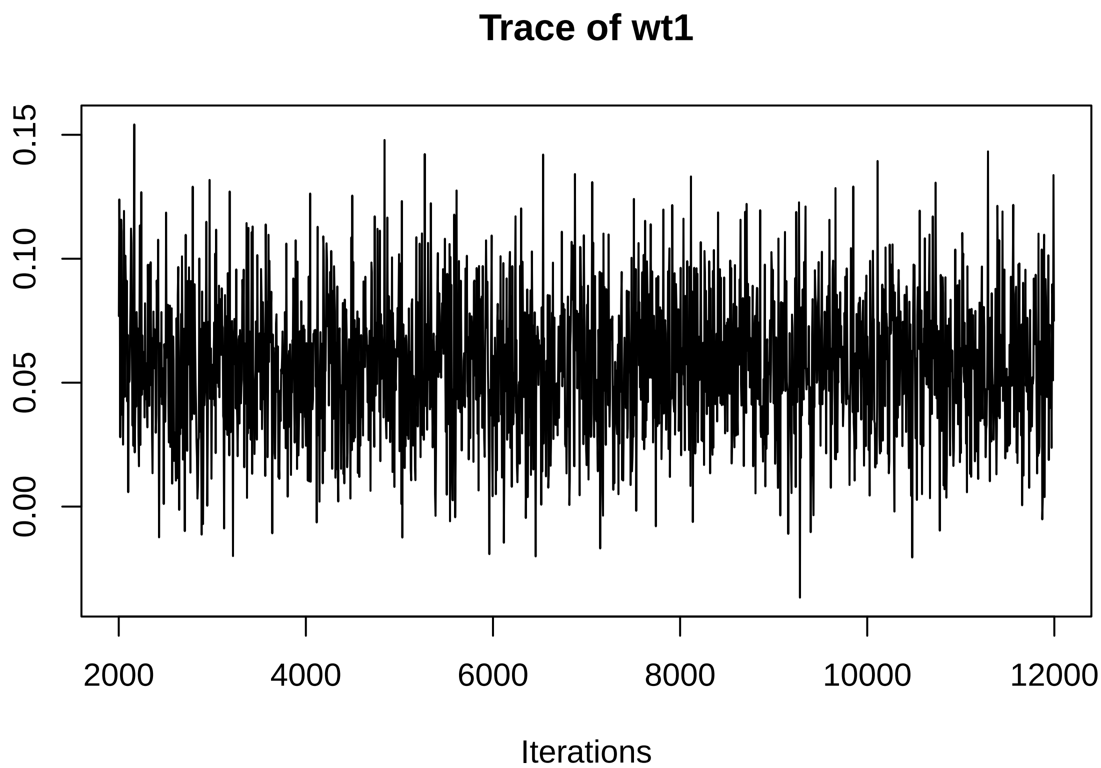


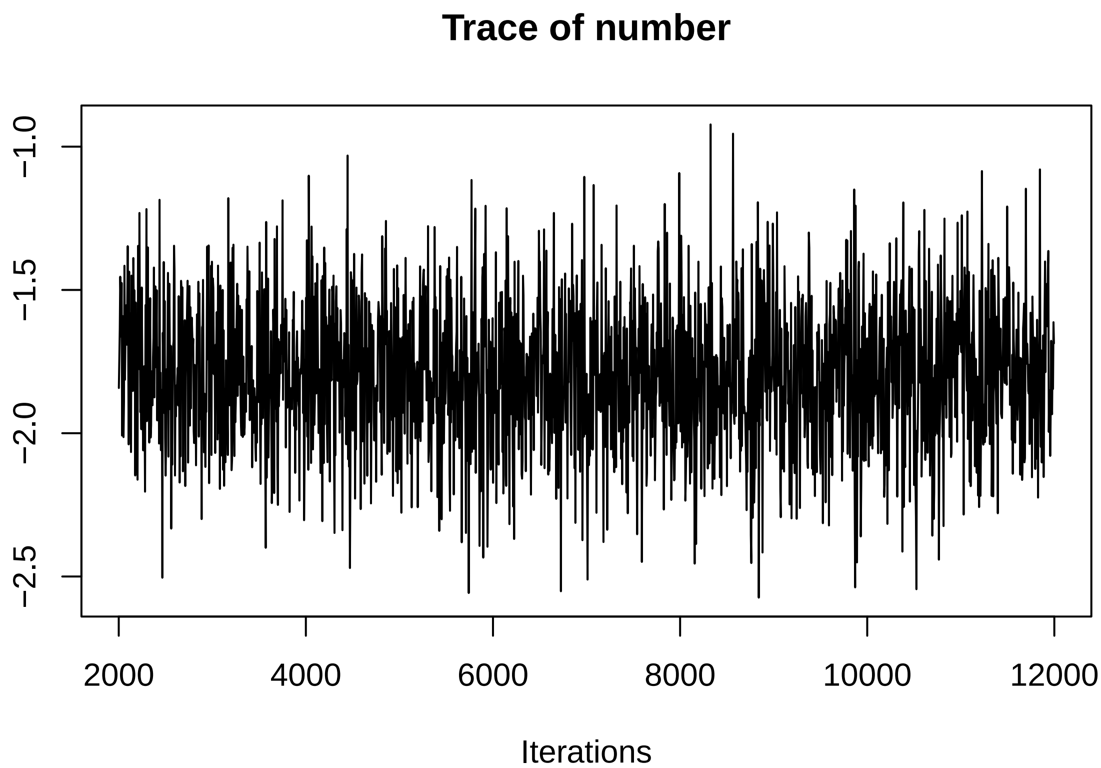

```r
traceplot(model)
```

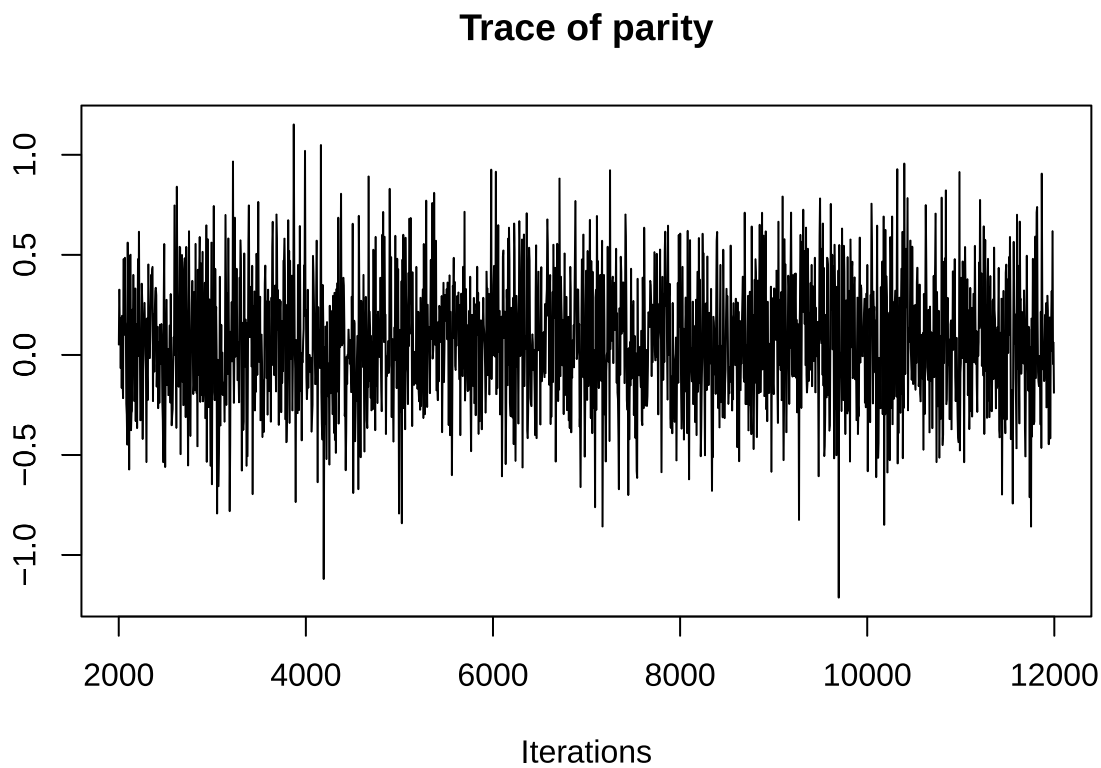

Trace of (Intercept)
−20
0
20
40
60
2000
4000
6000
8000
10000
12000
Iterations


Trace of gestation
0.03
0.02
0.01
−0.01
2000
4000
6000
8000
10000
12000
Iterations

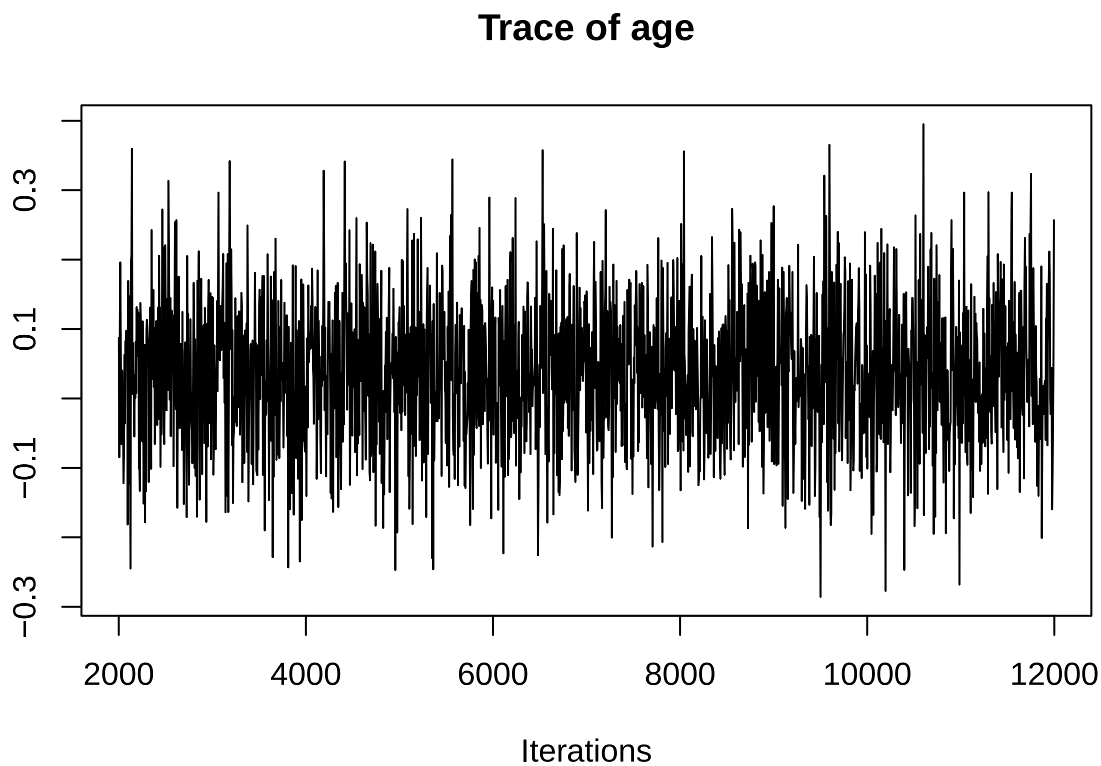

Trace of ht
0.5
1.0
1.5
2.0
2000
4000
6000
8000
10000
12000
Iterations

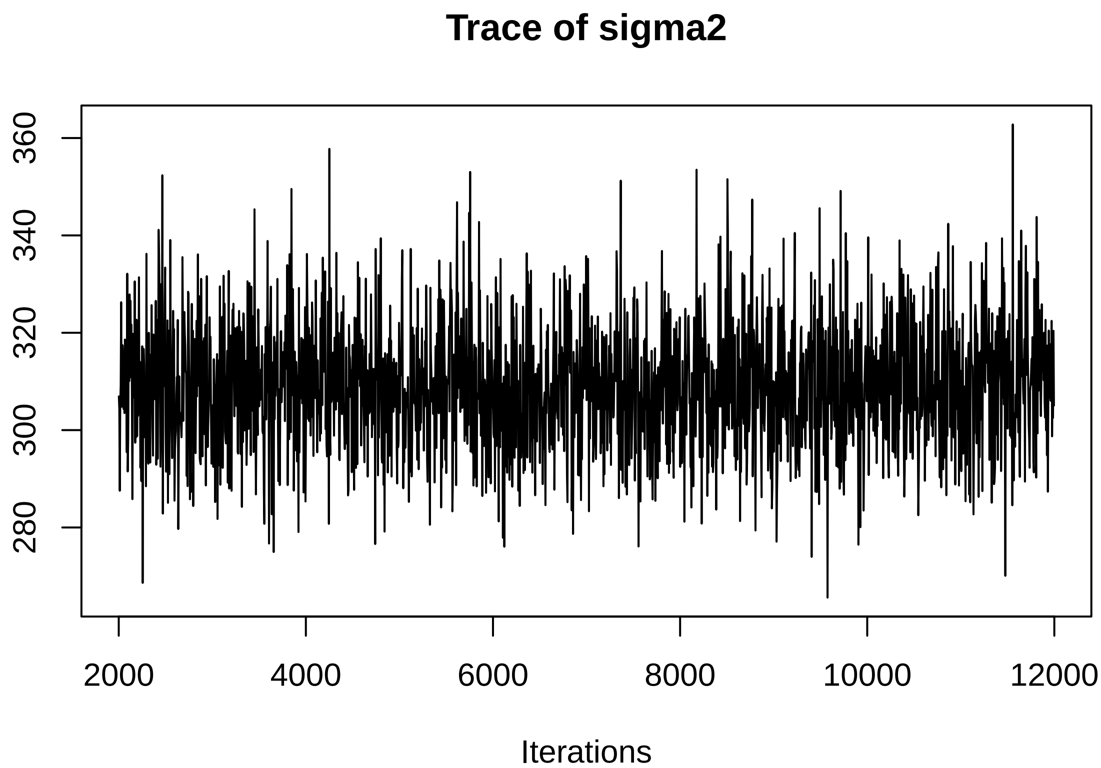

Trace of wt1
0.00
0.05
0.10
0.15
2000
4000
6000
8000
10000
12000
Iterations


Trace of number
−1.0
−1.5
−2.0
−2.5
2000
4000
6000
8000
10000
12000
Iterations

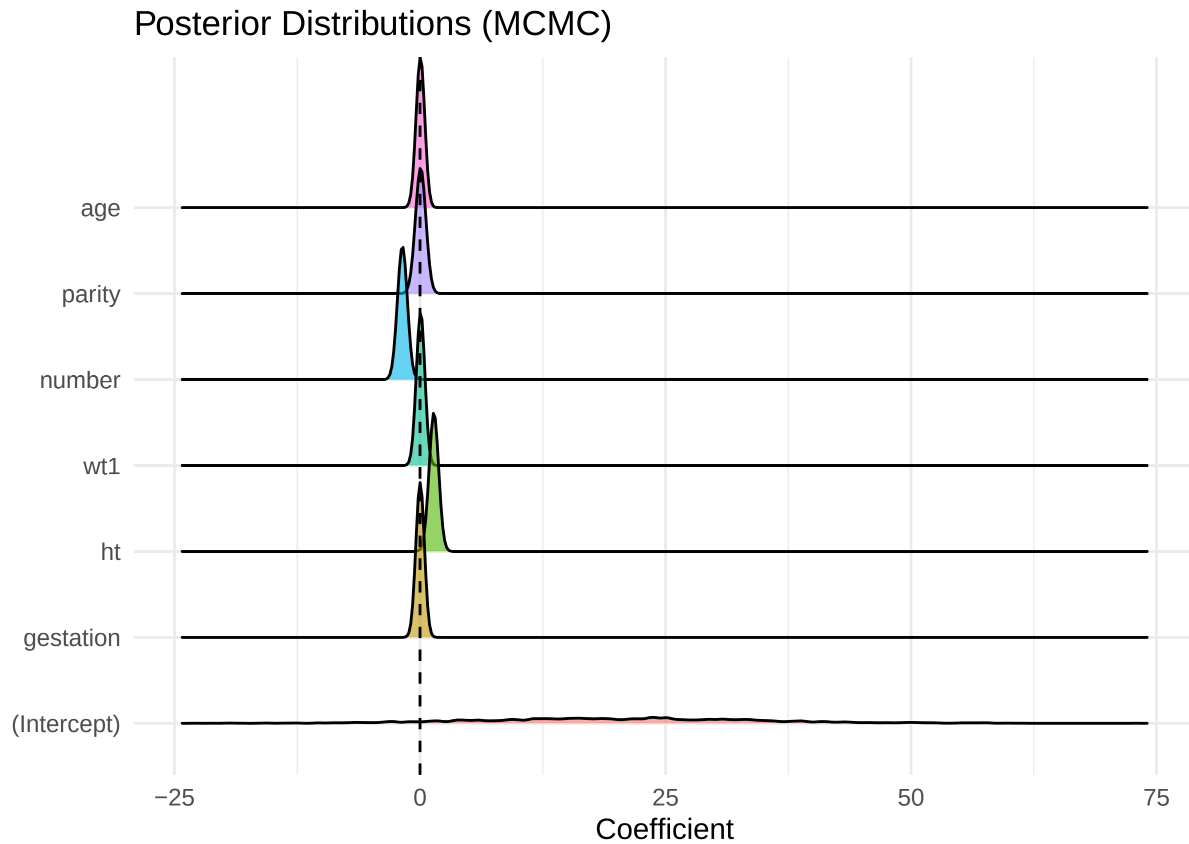


Trace of parity
1.0
0.5
0.0
−0.5
−1.0
2000
4000
6000
8000
10000
12000
Iterations


Trace of age
0.3
0.1
−0.1
−0.3
2000
4000
6000
8000
10000
12000
Iterations




```
#ridgeline plot
m <- as.matrix(model)
df <- as.data.frame(m)

#remove variance parameter
#only want to plot out coefficients
df <- df[, !colnames(df) %in% "sigma2"]

#convert to long format
df_long <- melt(df)

#plot
ggplot(df_long, aes(x = value, y = variable, fill = variable)) +
  geom_density_ridges(alpha = 0.6) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(
    title = "Posterior Distributions (MCMC)",
    x = "Coefficient",
    y = ""
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```



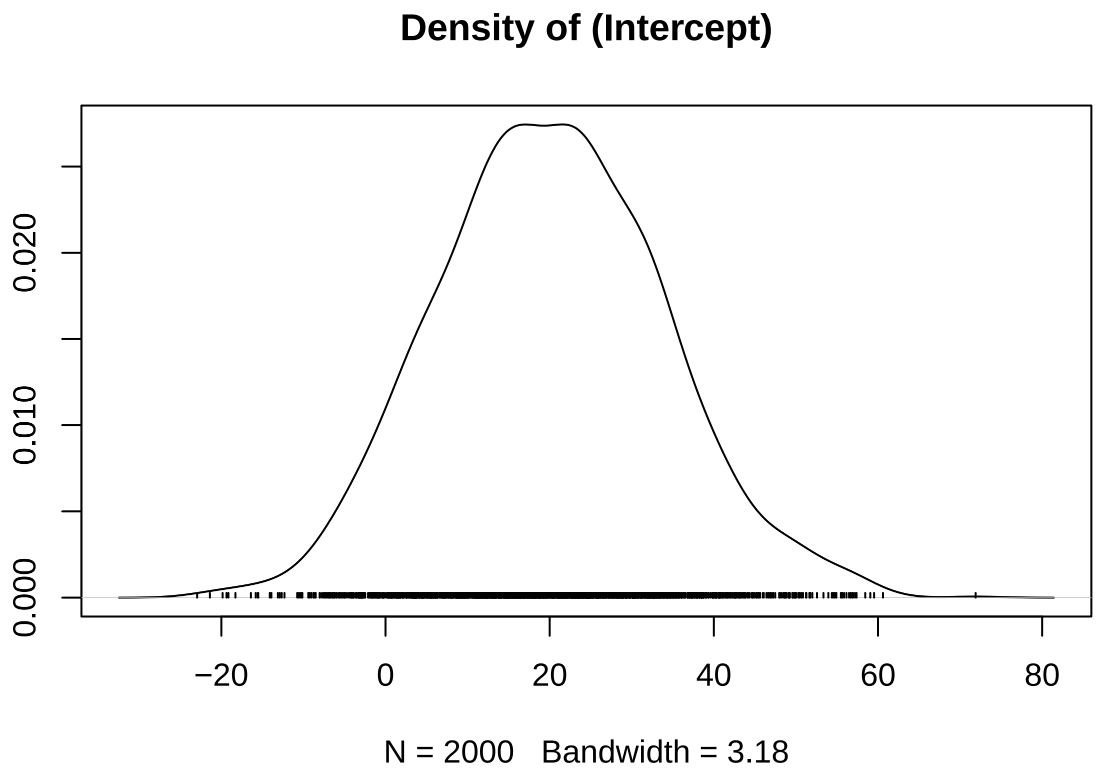
MCMC Informed Prior (Strong Means, Diffuse Variance)

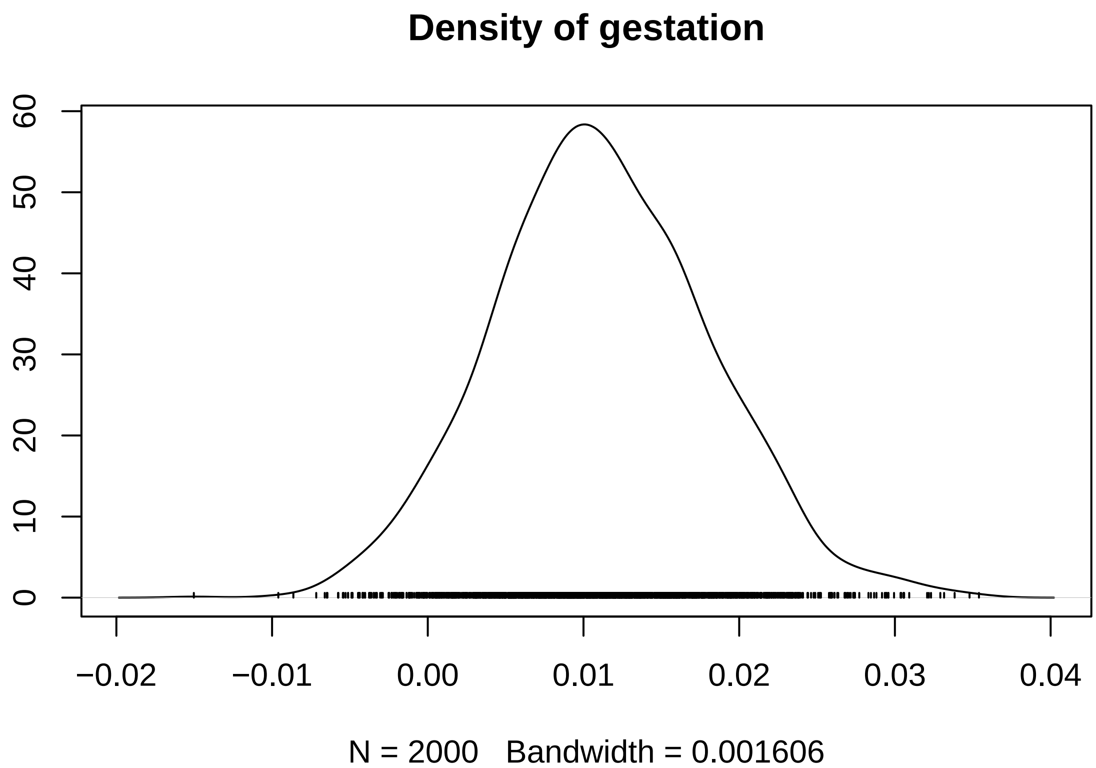
```
b0 <- c(
  0.00,    # intercept
  0.70,    # gestation
  0.84,    # ht
  0.12,    # wt1
  -1.80,   # number (cigs)
  3.5,    # parity
  0.00     # age
)

#precision matrix, required by MCMC pack
B0 <- diag(length(b0)) * (1 / 10000)

model_inf <- MCMCregress(
  wt ~ gestation + ht + wt1 + number + parity + age,
  data = babies_clean_2,
  b0 = b0,
  B0 = B0,
  burnin = 2000,
  mcmc = 10000,
  thin = 5
)
```

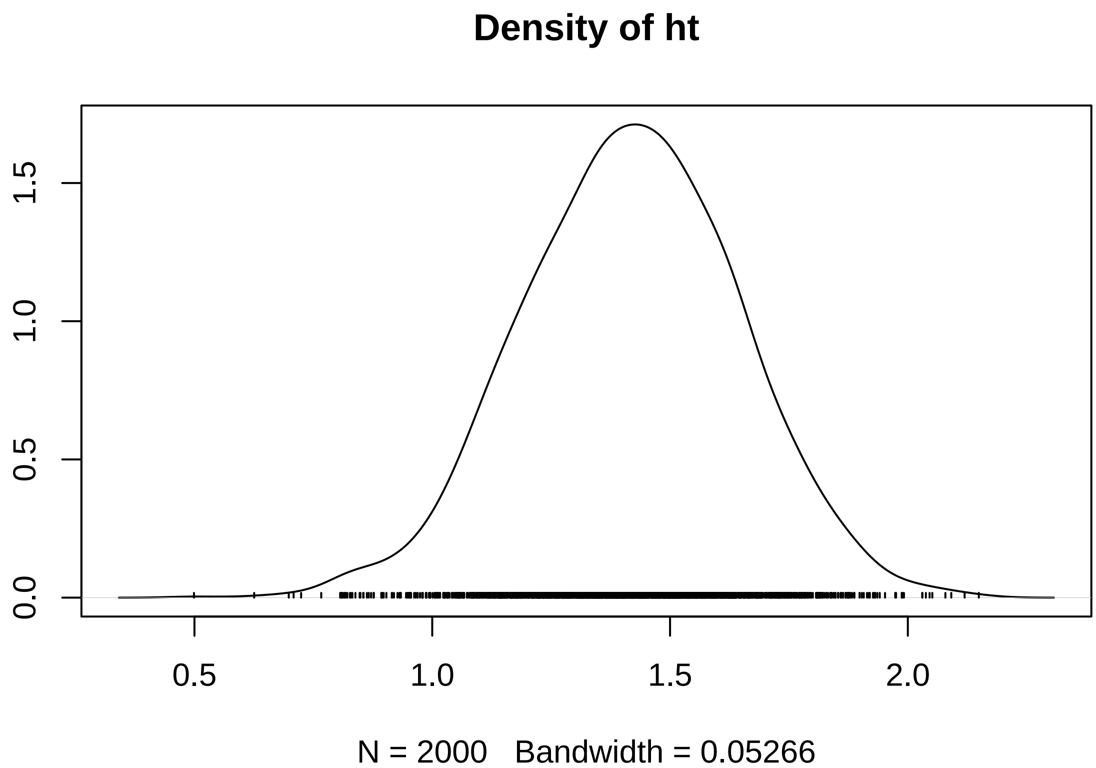
```r
summary(model_inf)
```

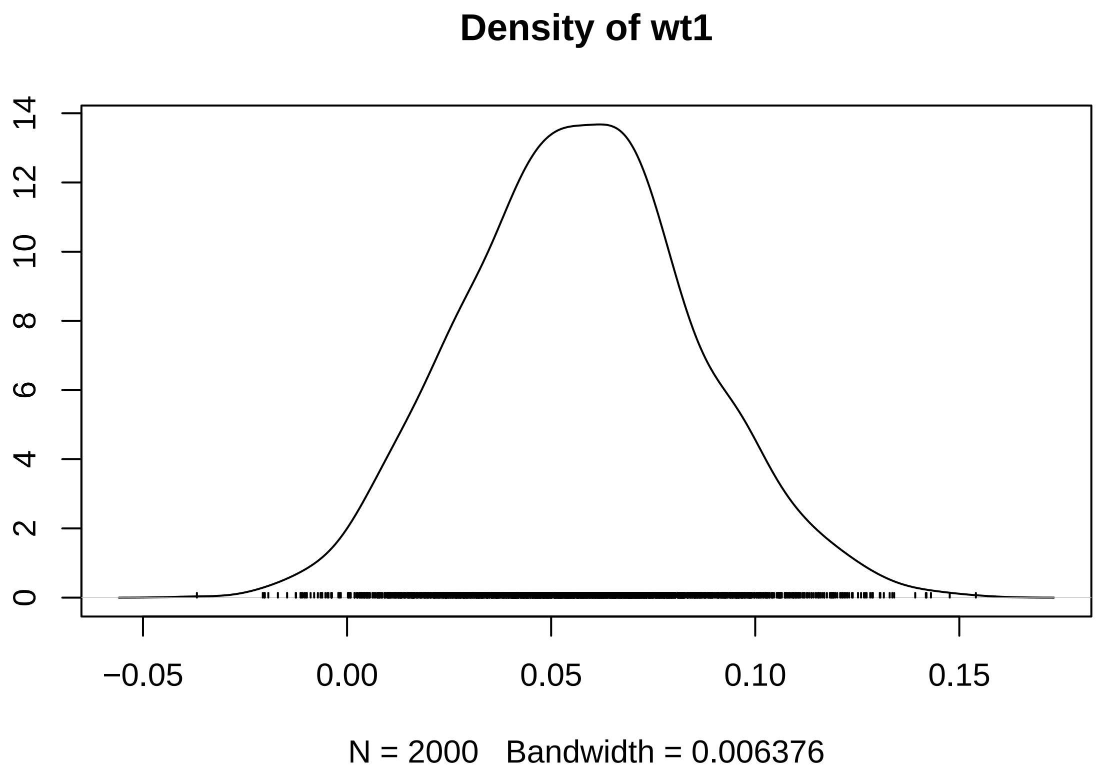
```
##
## Iterations = 2001:11996
## Thinning interval = 5
## Number of chains = 1
## Sample size per chain = 2000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##                  Mean       SD  Naive SE Time-series SE
## (Intercept)  20.08613 13.71998 0.3067880      0.2773522
## gestation     0.01115  0.00697 0.0001559      0.0001610
## ht            1.41638  0.22719 0.0050802      0.0046991
## wt1           0.05693  0.02829 0.0006327      0.0006327
## number       -1.79636  0.24848 0.0055561      0.0055561
## parity        0.06454  0.31299 0.0069986      0.0069986
## age           0.03808  0.10267 0.0022958      0.0022088
## sigma2      308.95440 12.96252 0.2898507      0.2898507
##
## 2. Quantiles for each variable:
##
##                   2.5%        25%       50%       75%     97.5%
## (Intercept)  -5.640166  10.906696  19.82416  29.46017  48.86751
## gestation    -0.002044   0.006396   0.01086   0.01568   0.02515
## ht            0.964447   1.260966   1.42091   1.57124   1.85349
## wt1           0.003765   0.038077   0.05658   0.07494   0.11476
## number       -2.284604  -1.957329  -1.79843  -1.63161  -1.31276
## parity       -0.535351  -0.142958   0.05904   0.26996   0.69140
## age          -0.162427  -0.030316   0.03739   0.10654   0.23756
## sigma2      285.334853 300.086056 308.63565 317.21774 335.73614
```

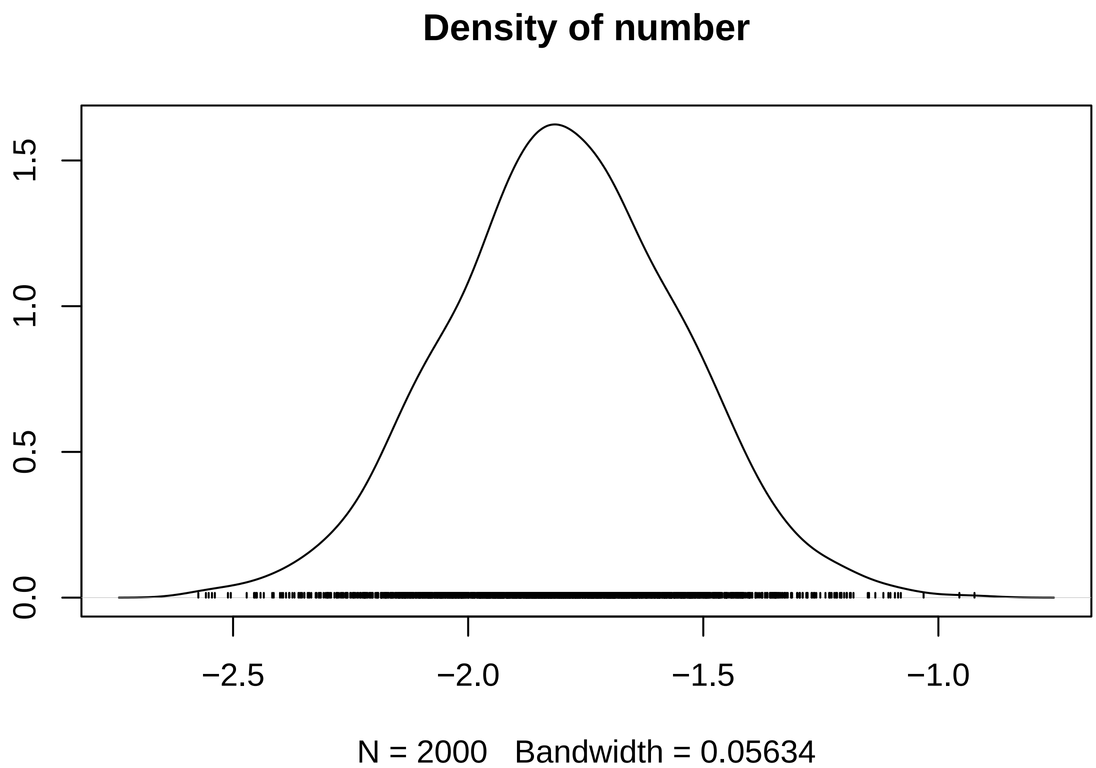
```r
densplot(model_inf)
```

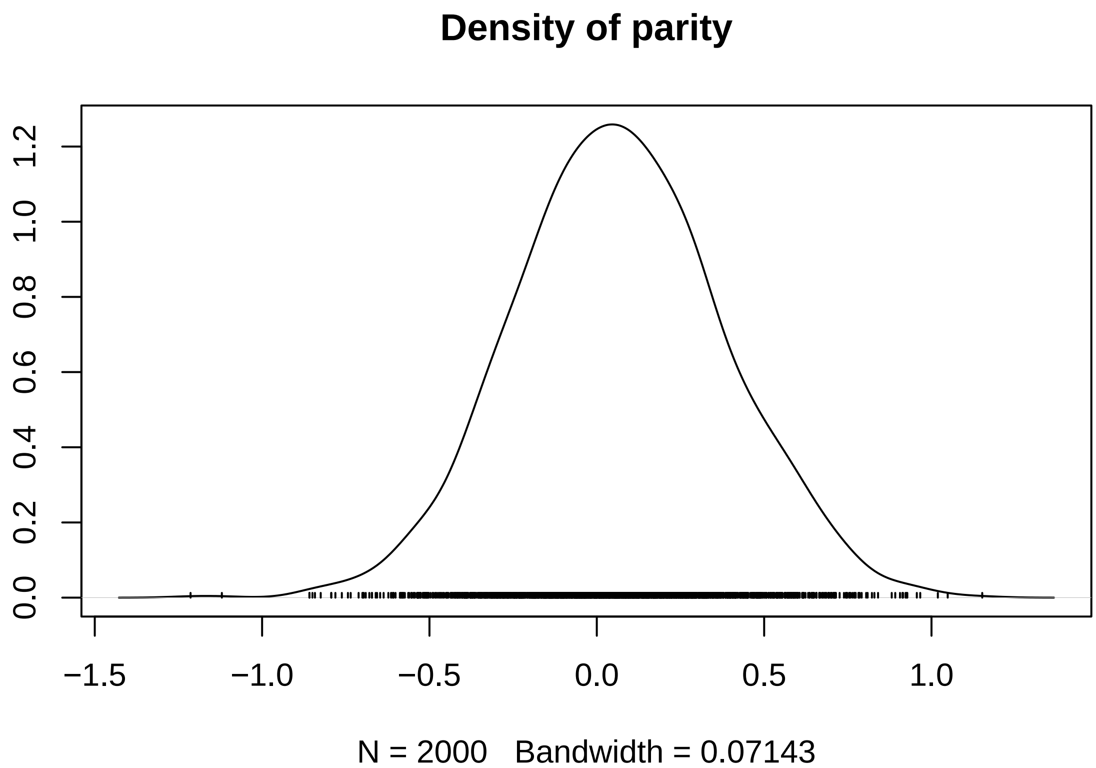

Density of (Intercept)
0.000
0.010
0.020
−20
0
20
40
60
80
N = 2000   Bandwidth = 3.18

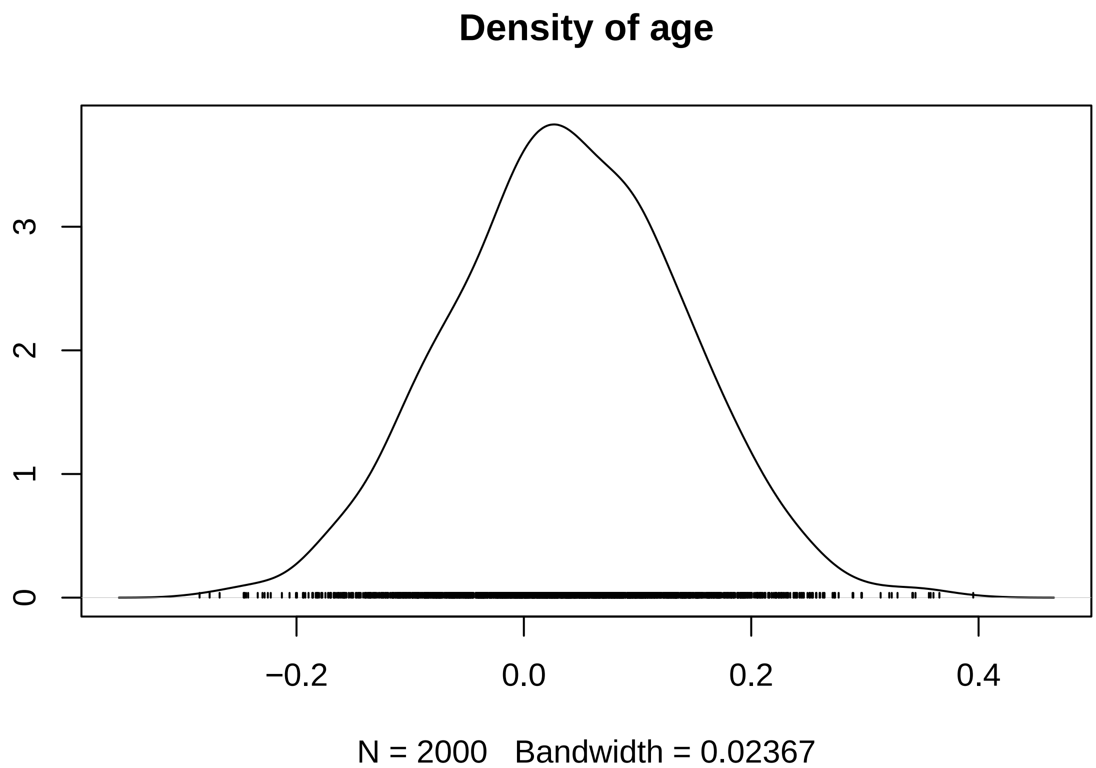

Density of gestation
60
50
40
30
20
10
0
−0.02
−0.01
0.00
0.01
0.02
0.03
0.04
N = 2000   Bandwidth = 0.001606


Density of ht
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
N = 2000 Bandwidth = 0.05266

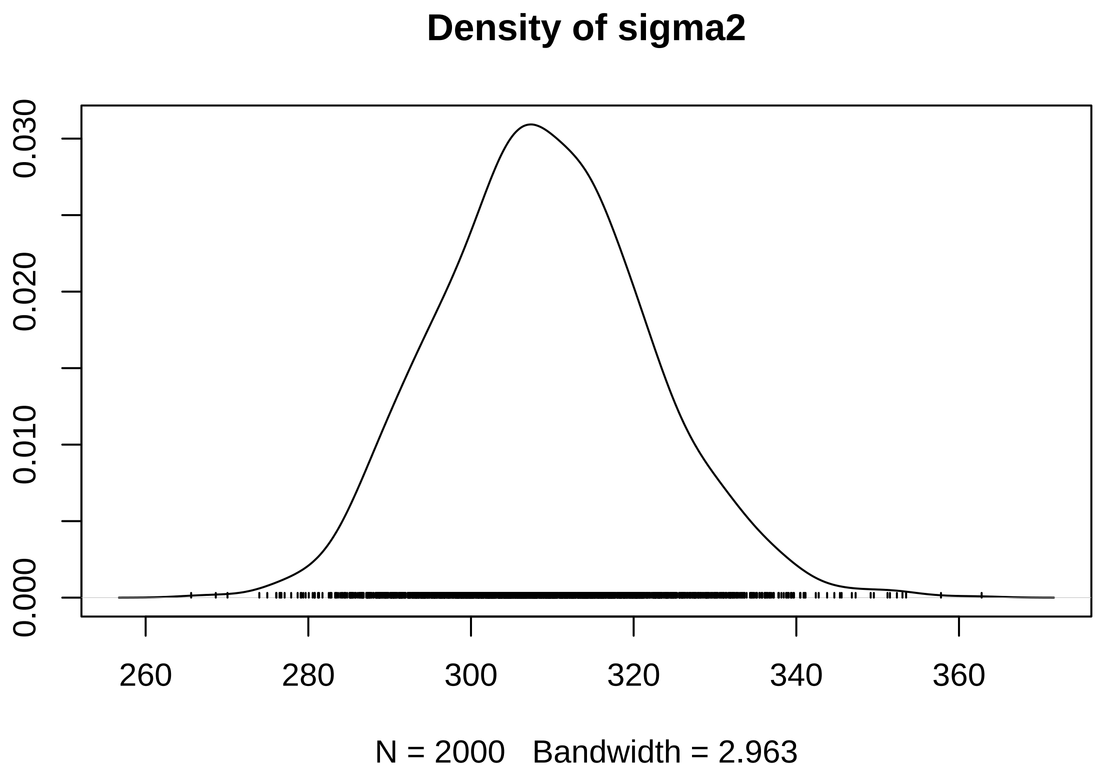

Density of wt1
0
2
4
6
8
10
12
14
−0.05
0.00
0.05
0.10
0.15
N = 2000   Bandwidth = 0.006376


Density of number
0.0
0.5
1.0
1.5
−2.5
−2.0
−1.5
−1.0
N = 2000 Bandwidth = 0.05634

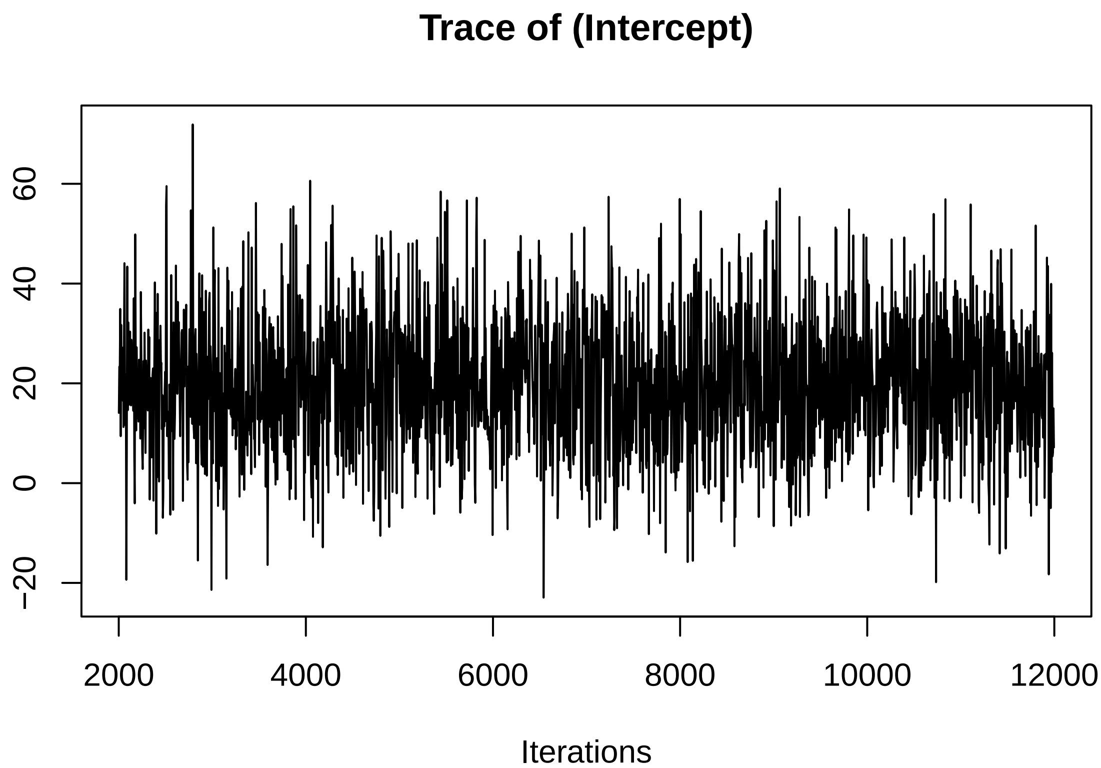

Density of parity
0.0
0.2
0.4
0.6
0.8
1.0
1.2
−1.5
−1.0
−0.5
0.0
0.5
1.0
N = 2000   Bandwidth = 0.07143



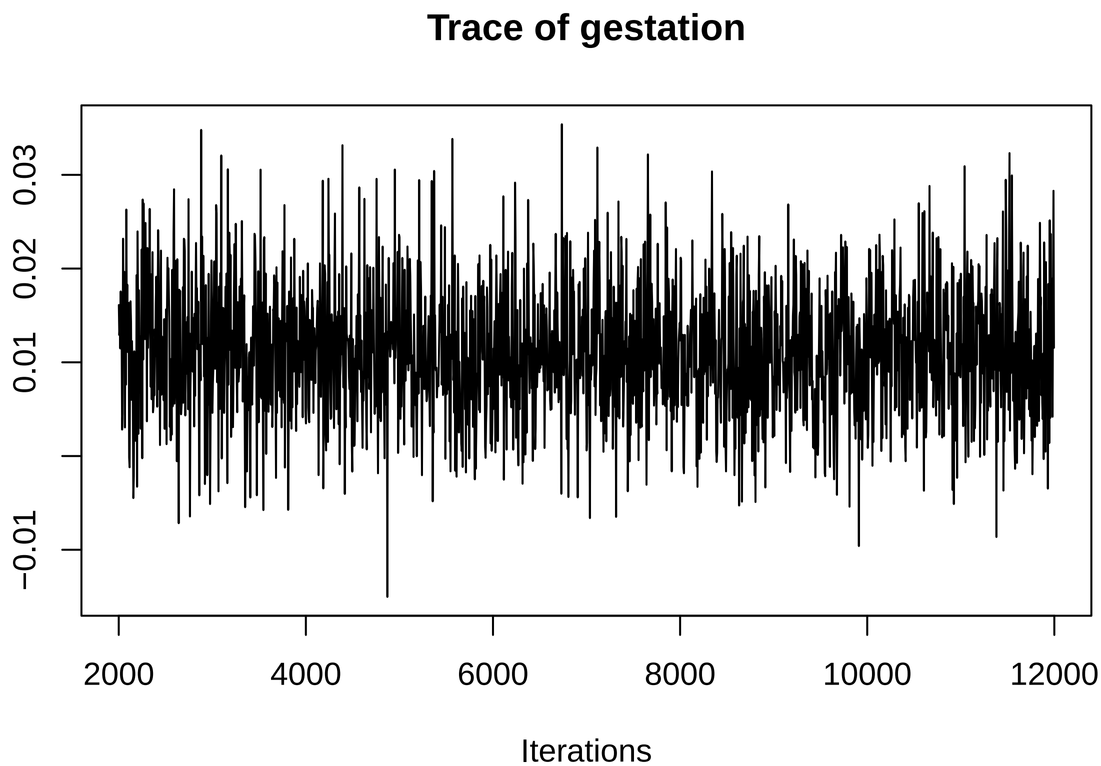
Density of age
3
2
1
0
−0.2
0.0
0.2
0.4
N = 2000   Bandwidth = 0.02367

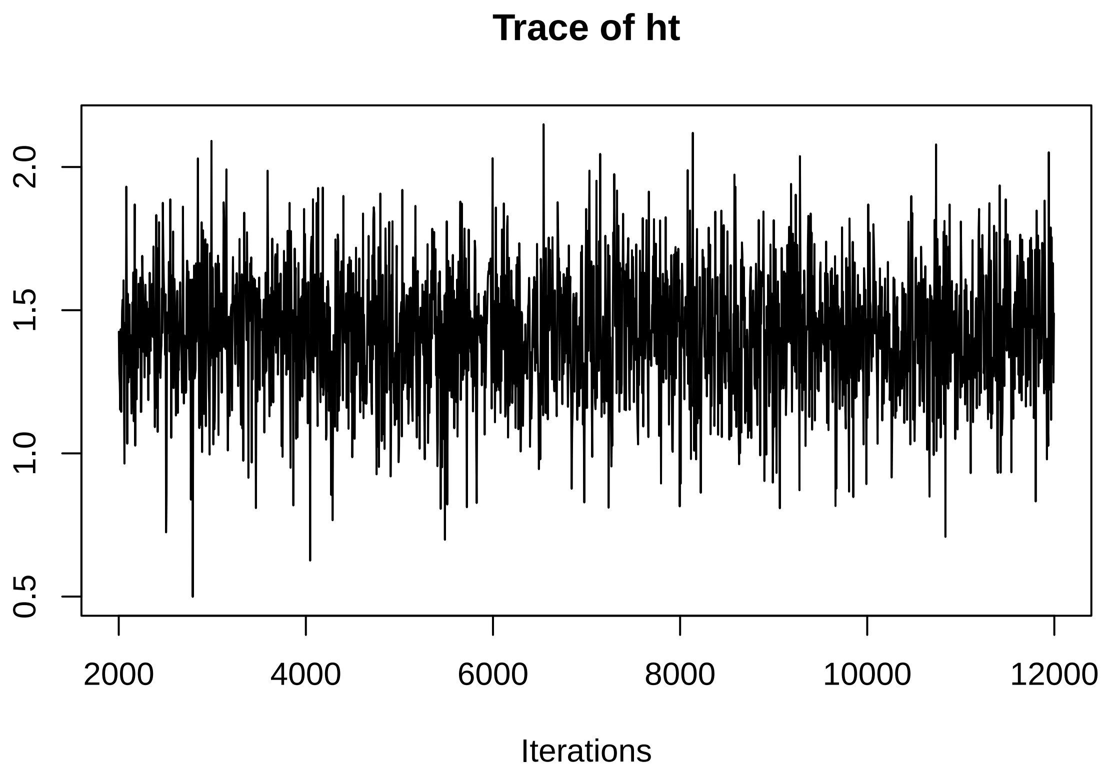



```r
traceplot(model_inf)
```

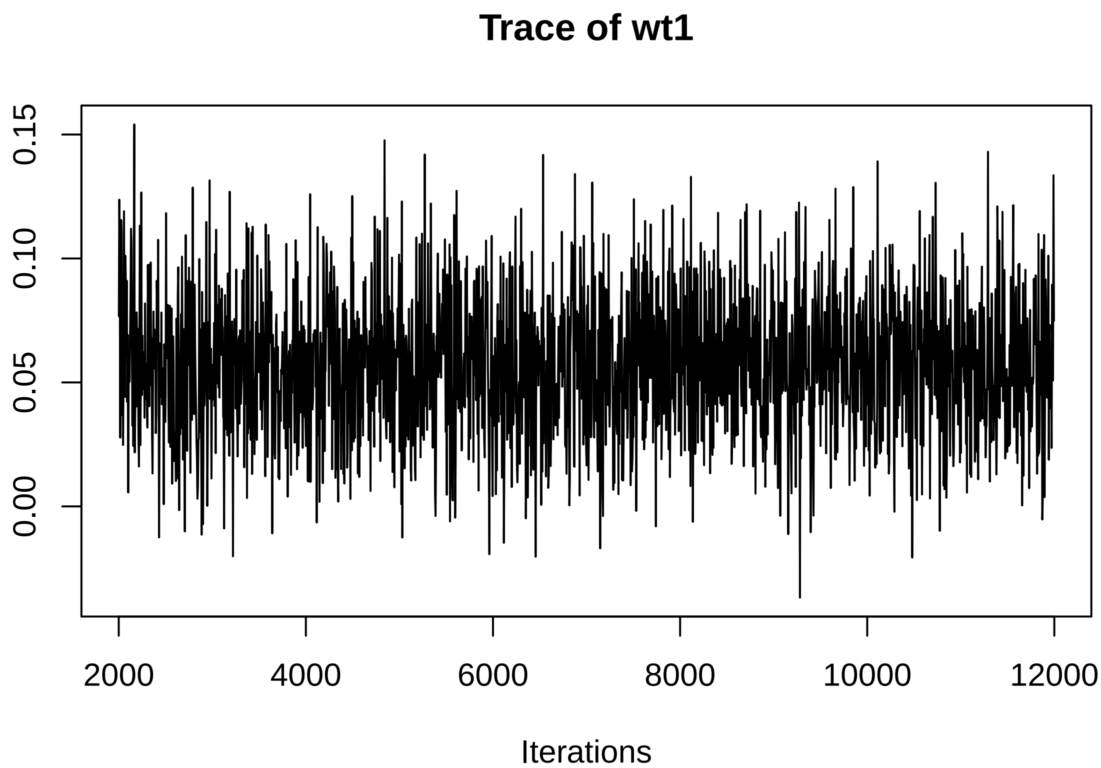


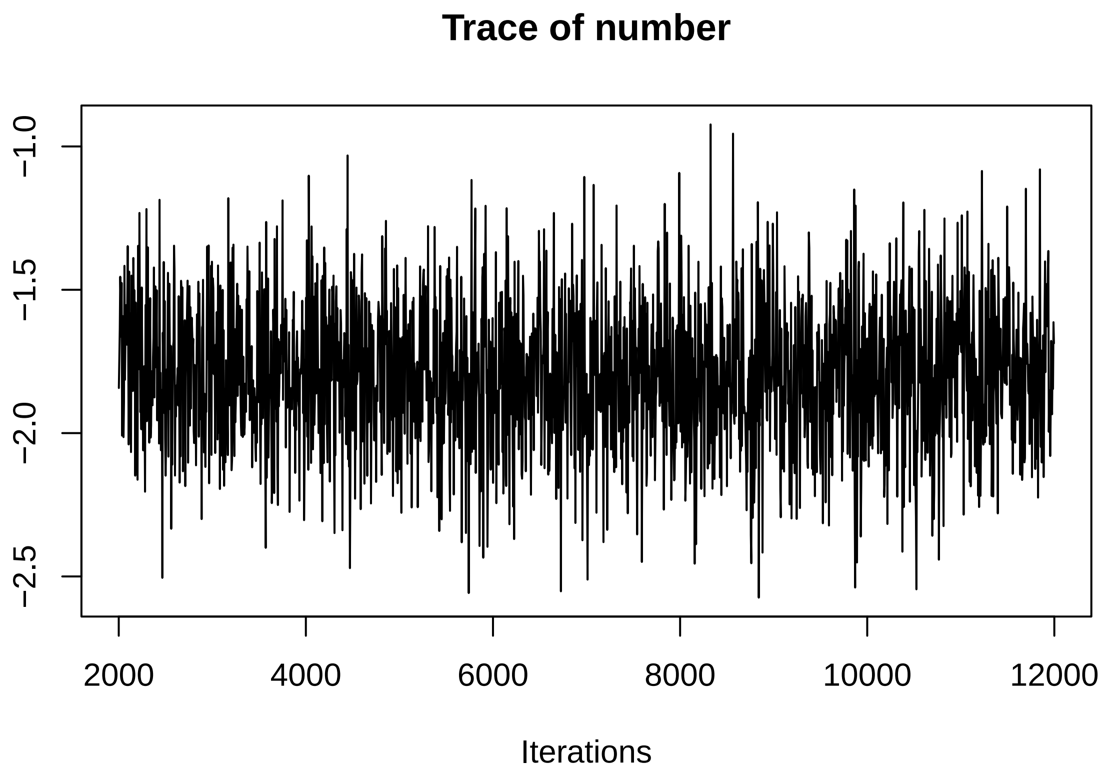
Trace of (Intercept)
60
40
20
0
−20
2000
4000
6000
8000
10000
12000
Iterations

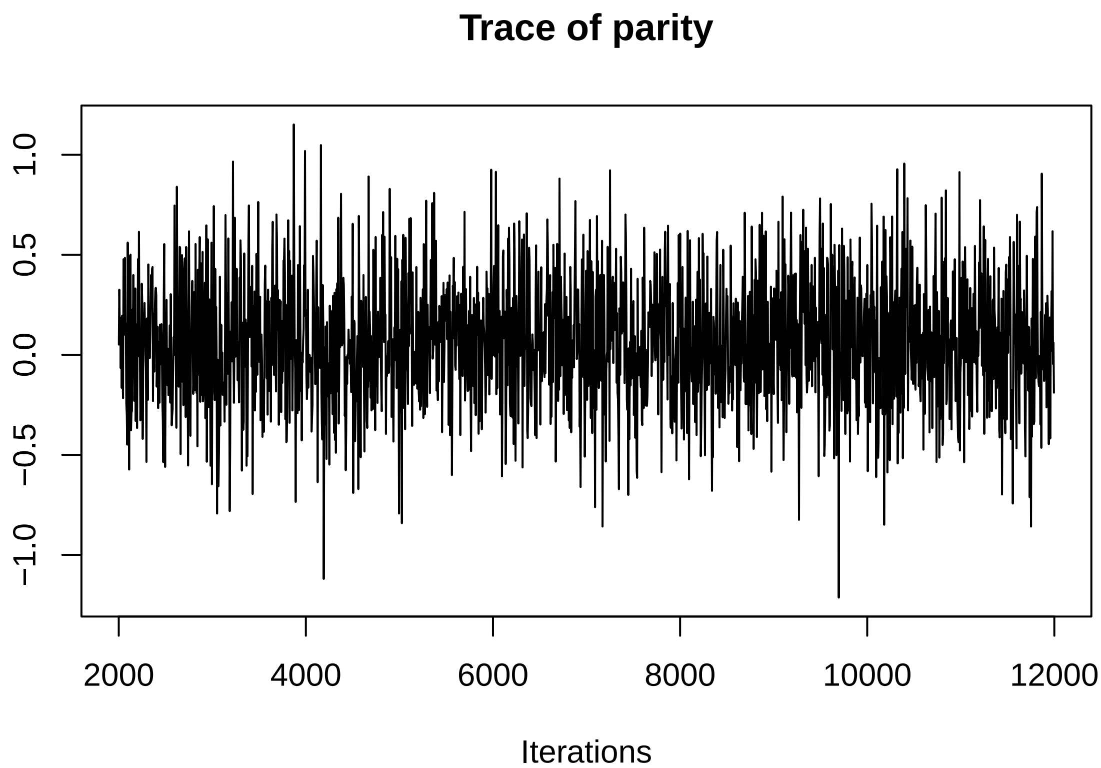

Trace of gestation
0.03
0.02
0.01
−0.01
2000
4000
6000
8000
10000
12000
Iterations

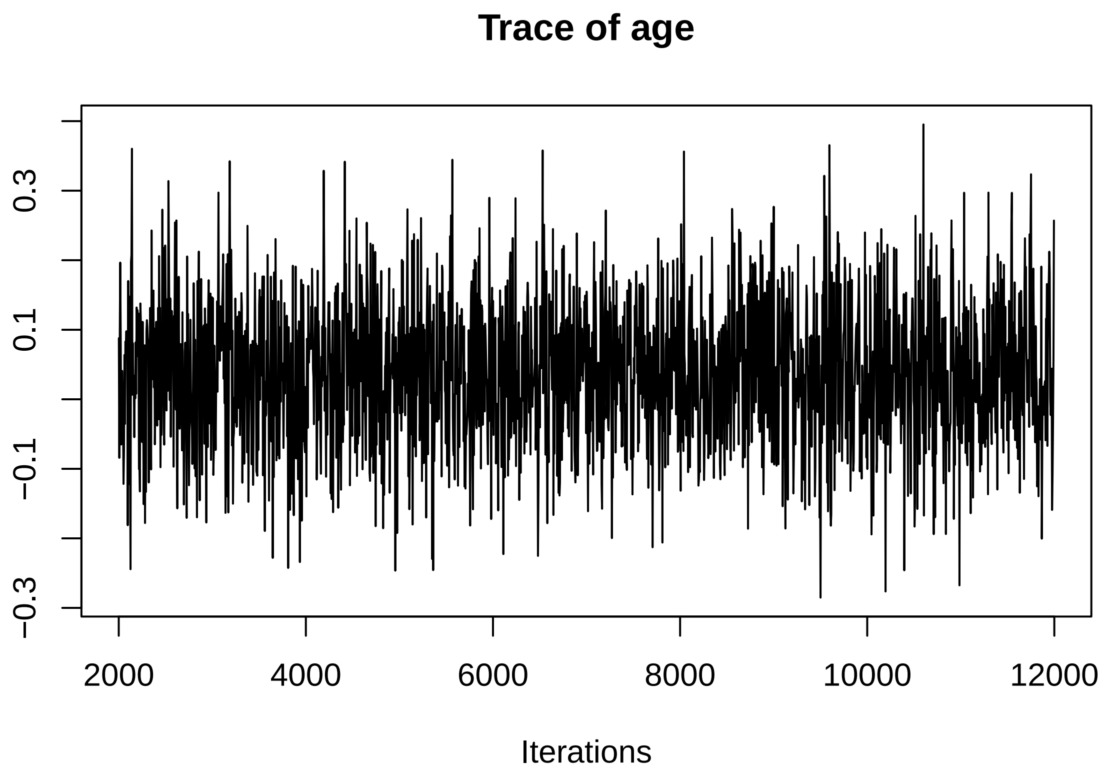


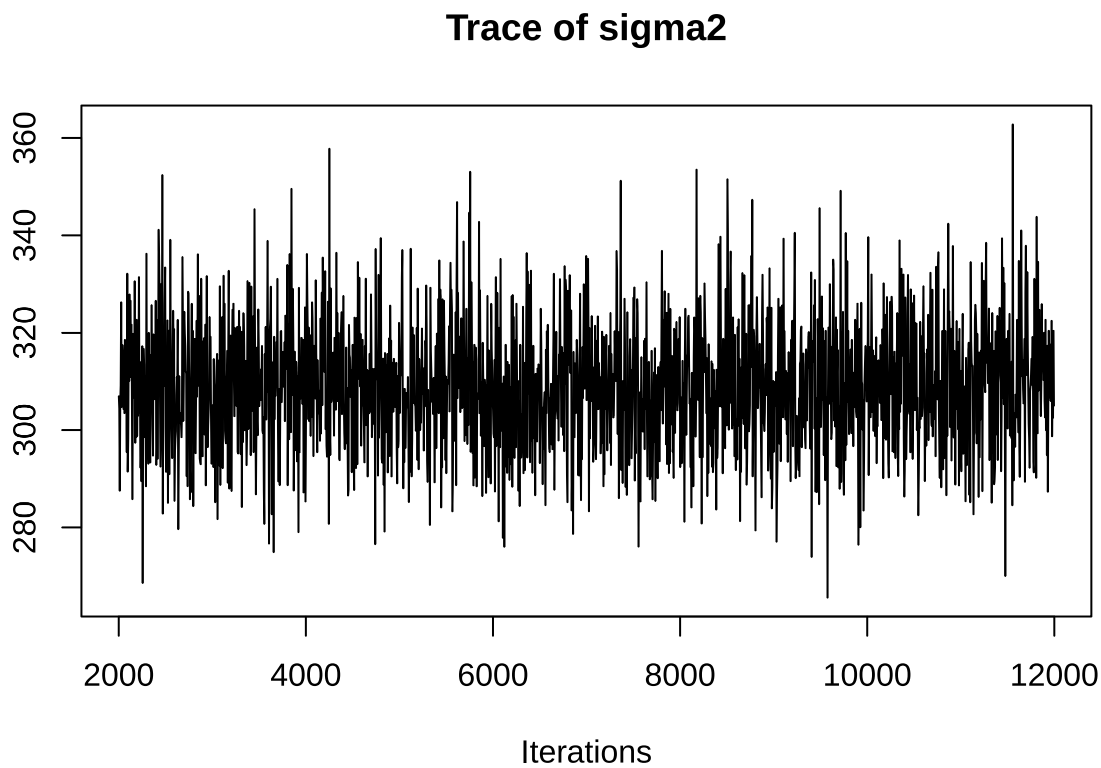
Trace of ht
0.5
1.0
1.5
2.0
2000
4000
6000
8000
10000
12000
Iterations


Trace of wt1
0.00
0.05
0.10
0.15
2000
4000
6000
8000
10000
12000
Iterations

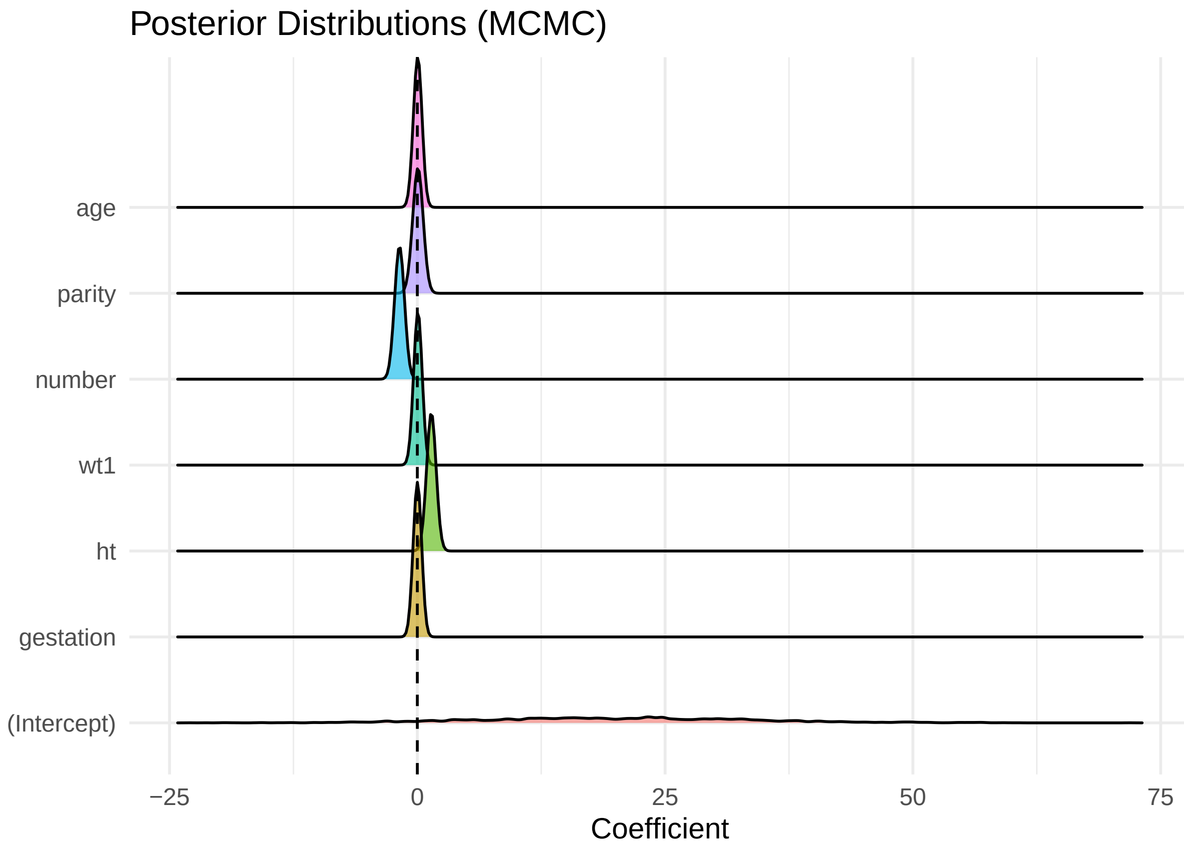

Trace of number
−1.0
−1.5
−2.0
−2.5
2000
4000
6000
8000
10000
12000
Iterations


Trace of parity
1.0
0.5
0.0
−0.5
−1.0
2000
4000
6000
8000
10000
12000
Iterations


Trace of age
0.3
0.1
−0.1
−0.3
2000
4000
6000
8000
10000
12000
Iterations




```r
#ridgeline plot
m2 <- as.matrix(model_inf)
df2 <- as.data.frame(m2)

#remove variance parameter
df2 <- df2[, !colnames(df2) %in% "sigma2"]

#convert to long format
df_long2 <- melt(df2)

#plot
ggplot(df_long2, aes(x = value, y = variable, fill = variable)) +
  geom_density_ridges(alpha = 0.6) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(
    title = "Posterior Distributions (MCMC)",
    x = "Coefficient",
    y = ""
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```


Posterior Distributions (MCMC)
age
parity
number
wt1
ht
gestation
(Intercept)
−25
0
25
50
75
Coefficient